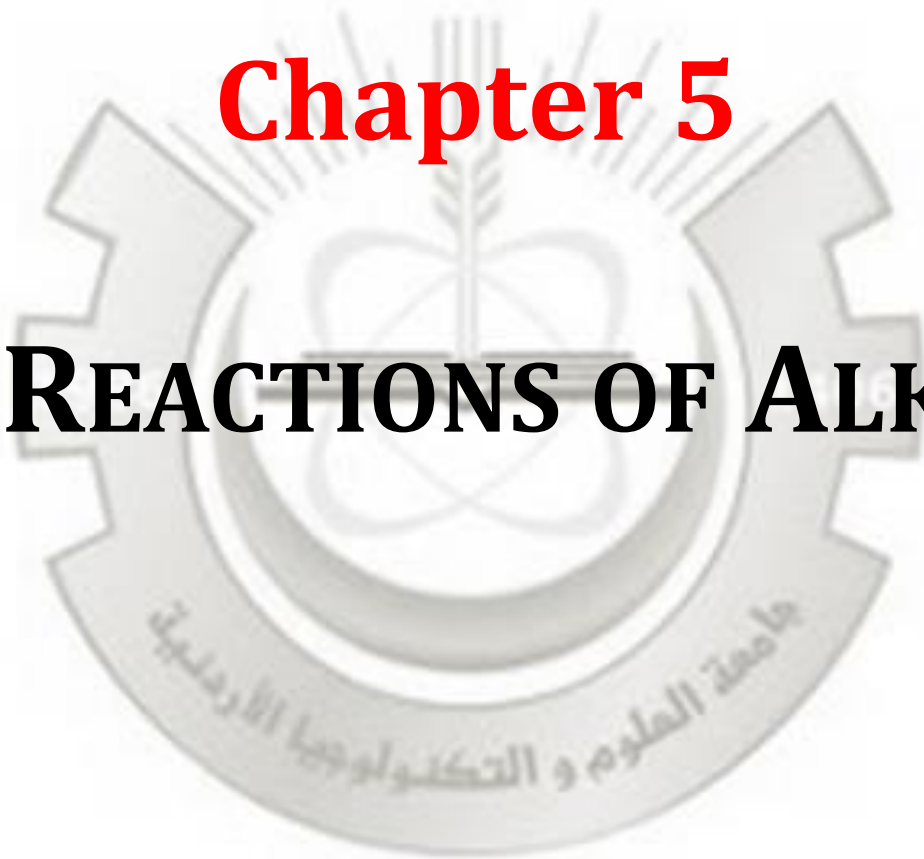


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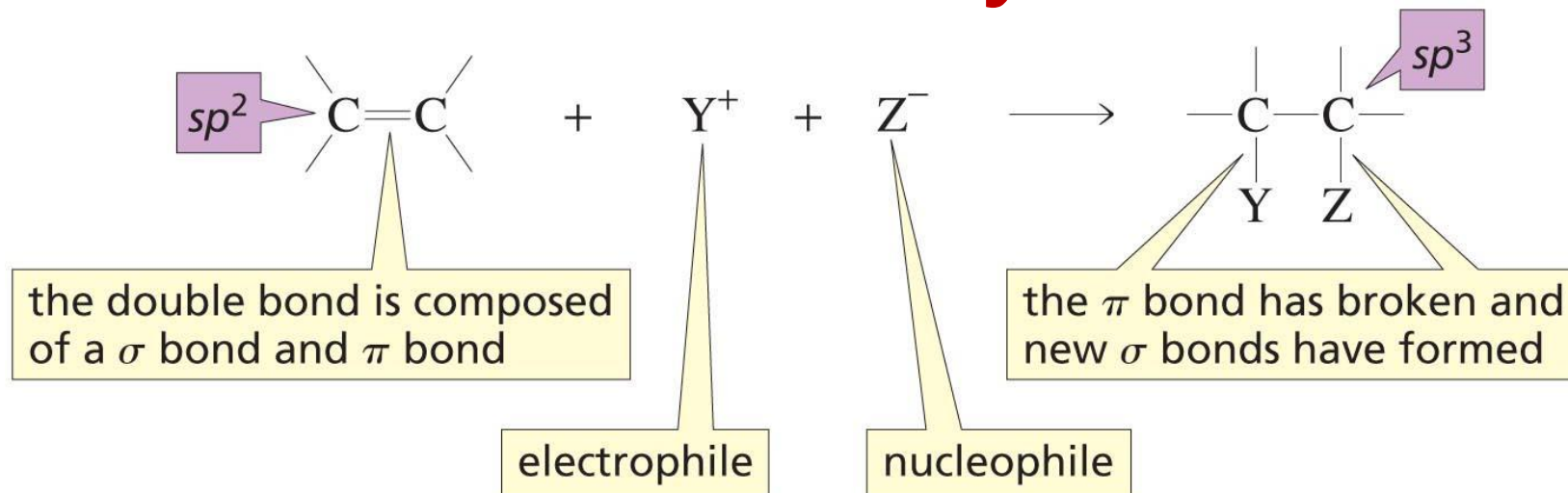
PAULA YURKANIS BRUCE

Chapter 5

THE REACTIONS OF ALKENES



Electrophilic Addition of Alkenes: Reaction and Synthesis



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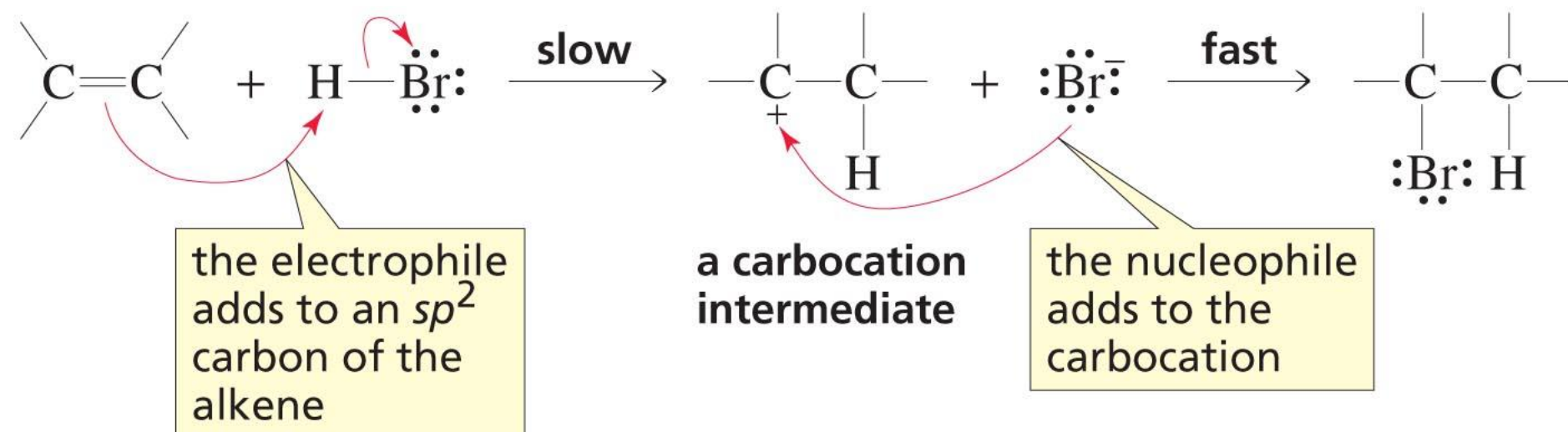
**Y-Z: H-Cl, H-Br, H-I,
Cl-Cl, & Br-Br**

**Reaction of an alkene is
an addition reaction.**



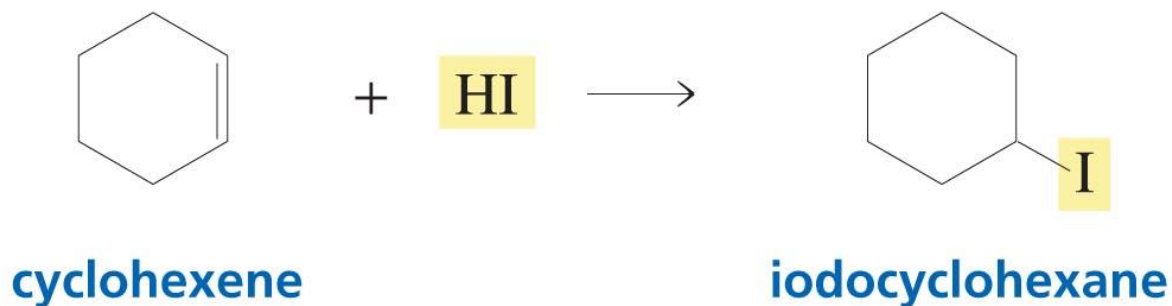
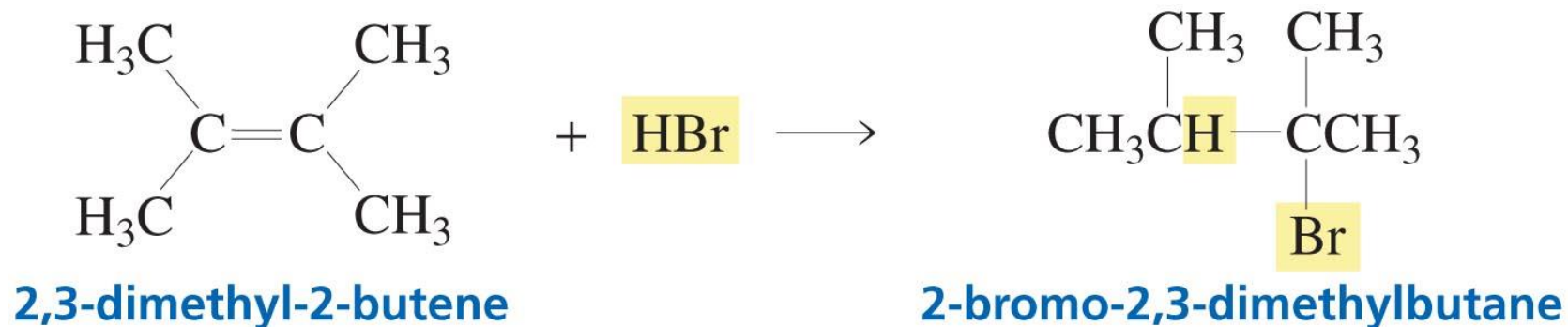
**synthesis of an alkene
an elimination reaction**

Electrophilic Addition of Alkenes



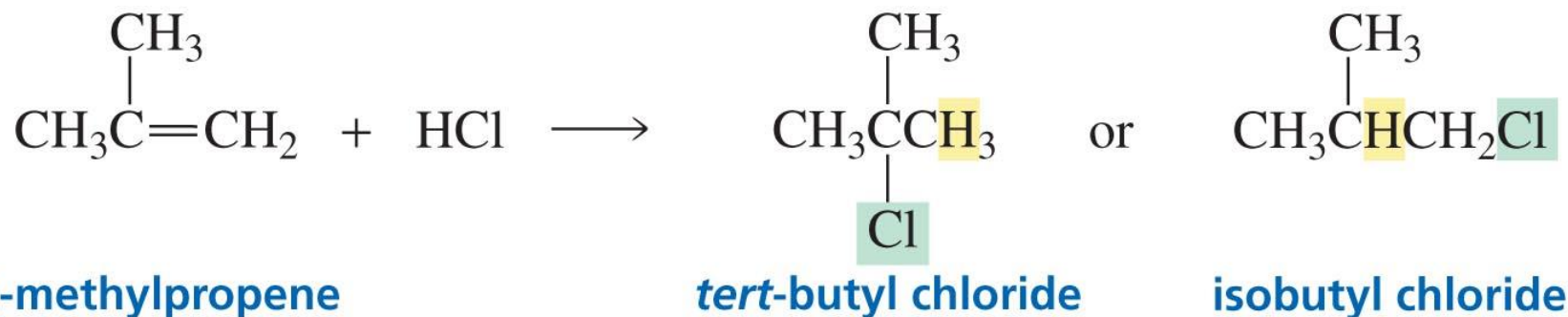
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Addition of Hydrogen Halides



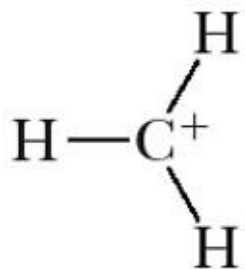
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What is the product?

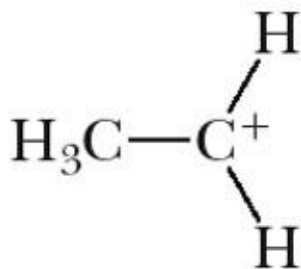


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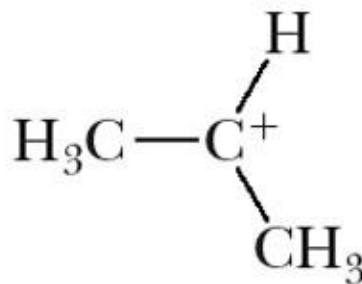
Carbocation Stabilities



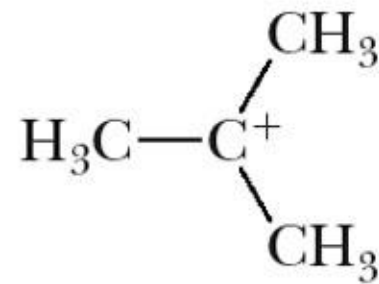
Methyl cation
(Methyl)



Ethyl cation
(1°)



Isopropyl cation
(2°)



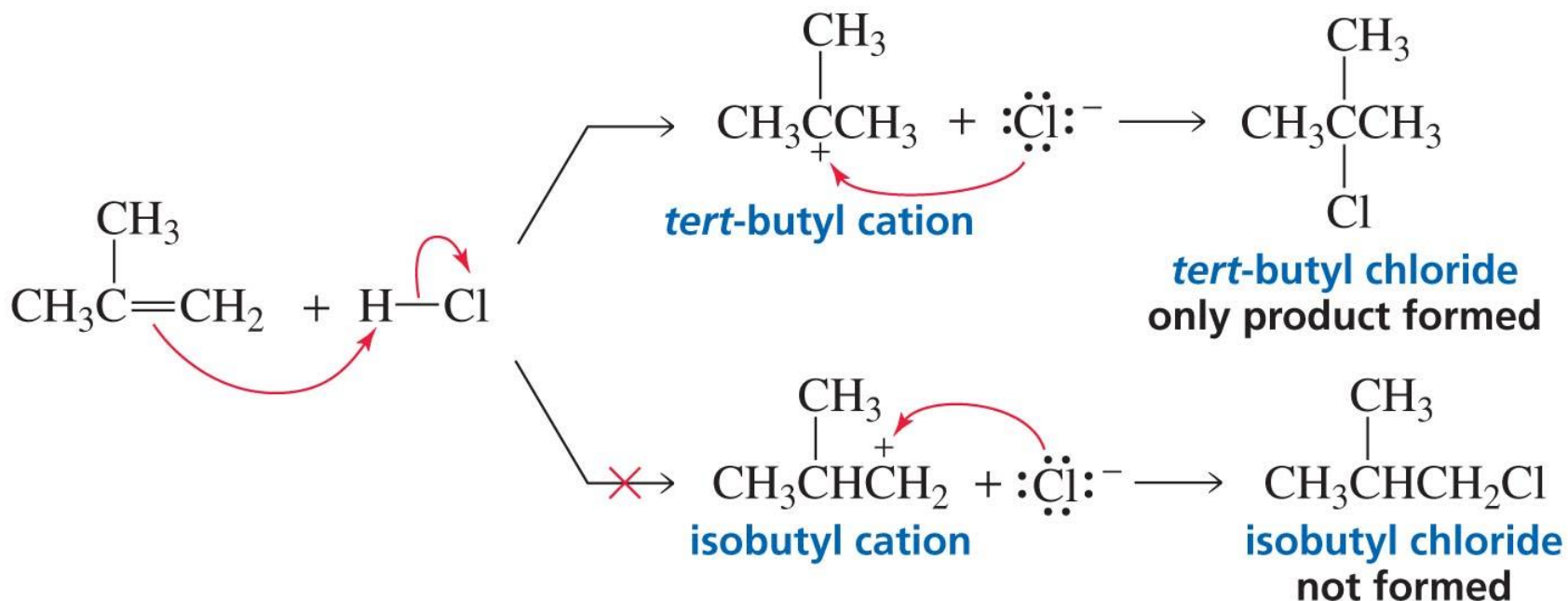
tert-Butyl cation
(3°)

Increasing carbocation stability

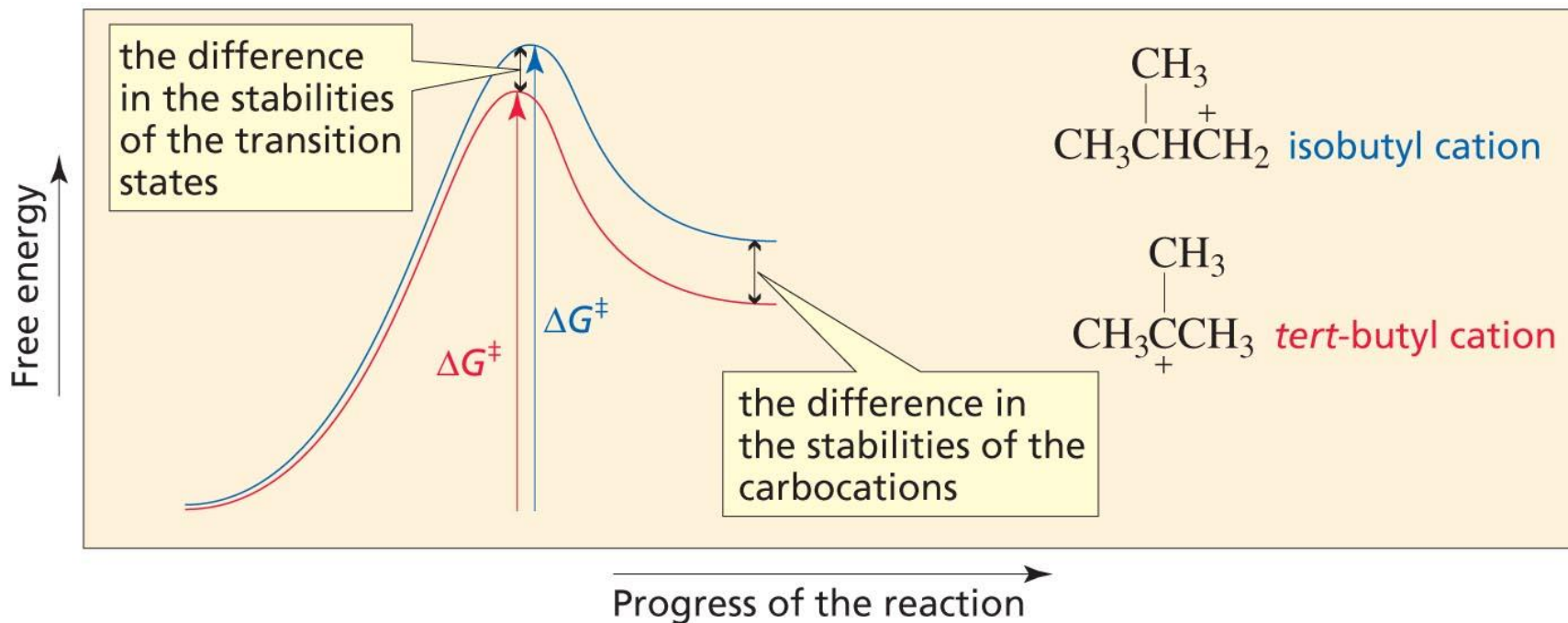
Increasing Ease of Formation

Carbocation Formation is the Rate-Limiting Step

**more stable
carbocation**



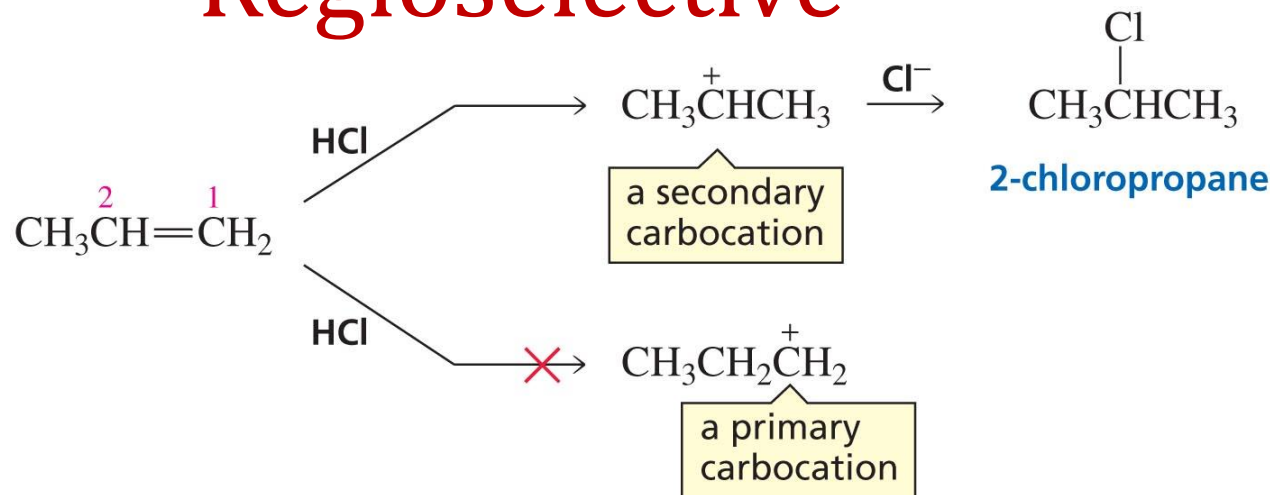
Reaction Coordinate Diagram for the Addition of H^+ to 2-Methylpropene



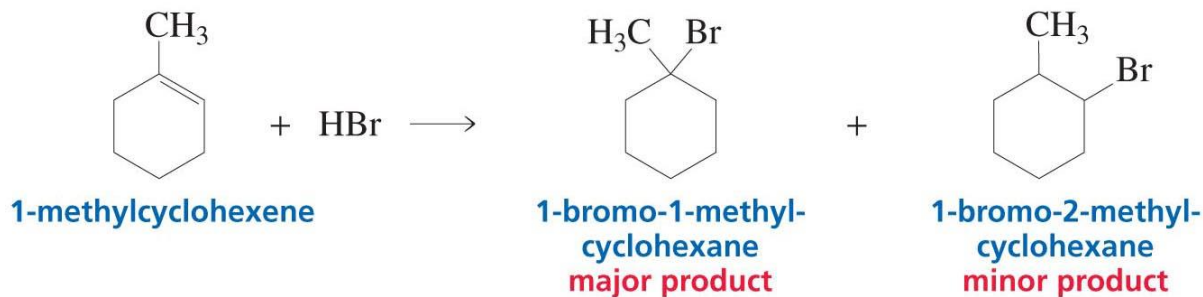
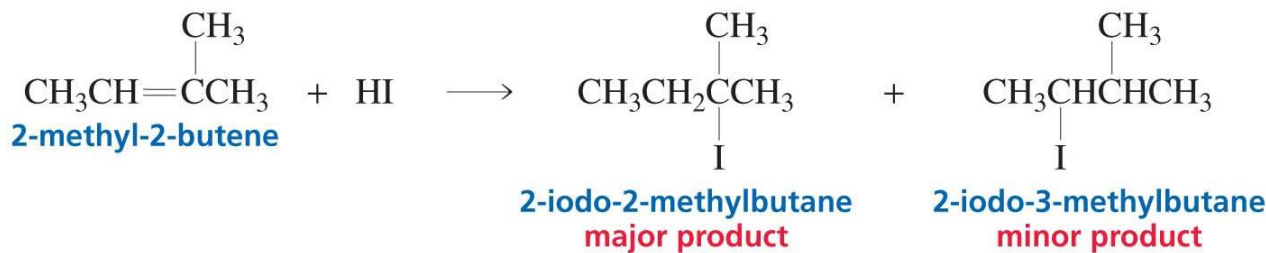
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In an electrophilic addition reaction, the more stable carbocation forms faster.

Electrophilic Addition Reactions are Regioselective



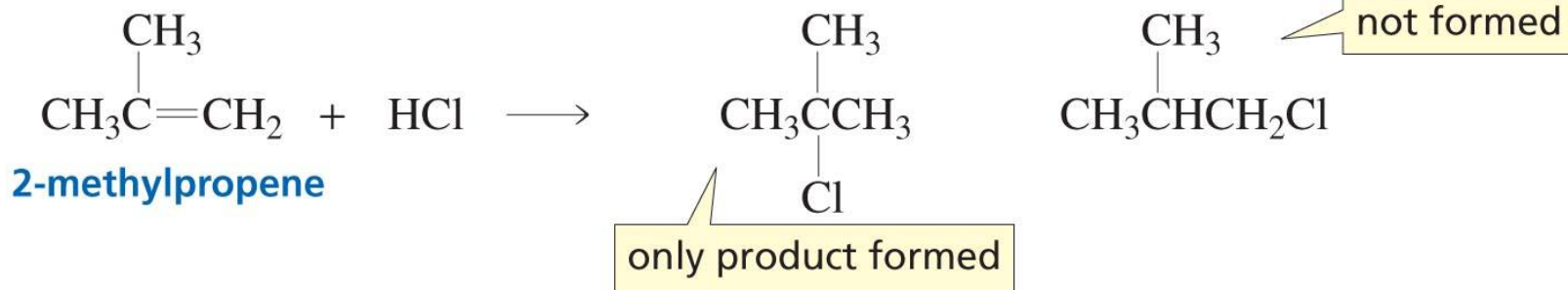
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Markovnikov's Rule

The electrophile adds to the sp^2 carbon that is bonded to the greater number of hydrogens:

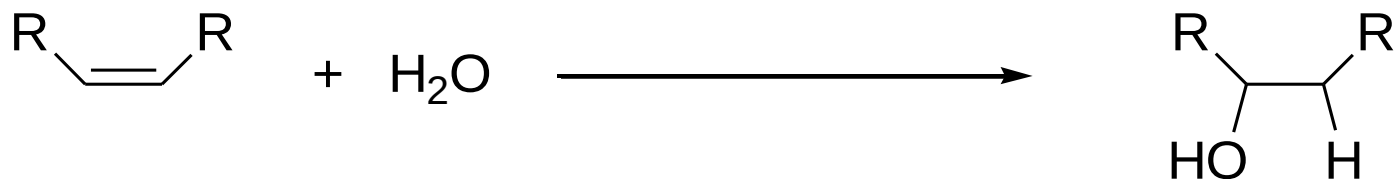


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In a regioselective reaction, one constitutional isomer is the major or the only product.

Addition of Water to Alkene

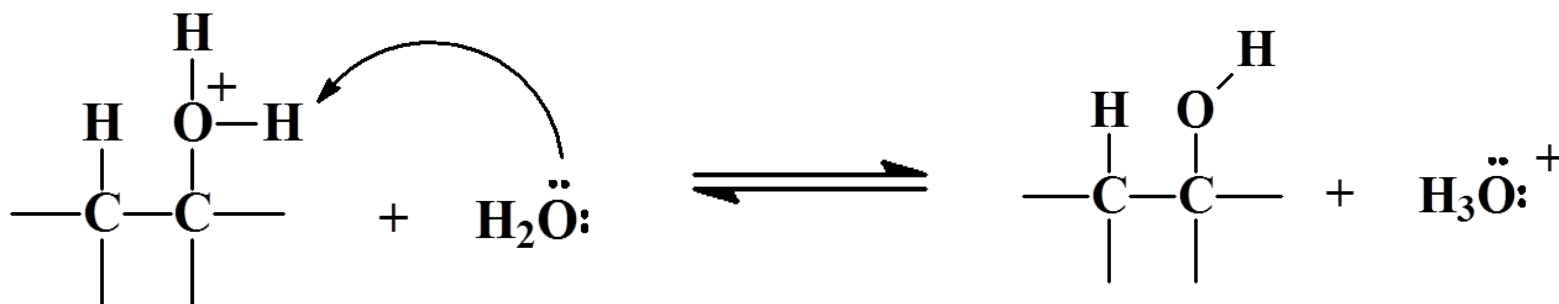
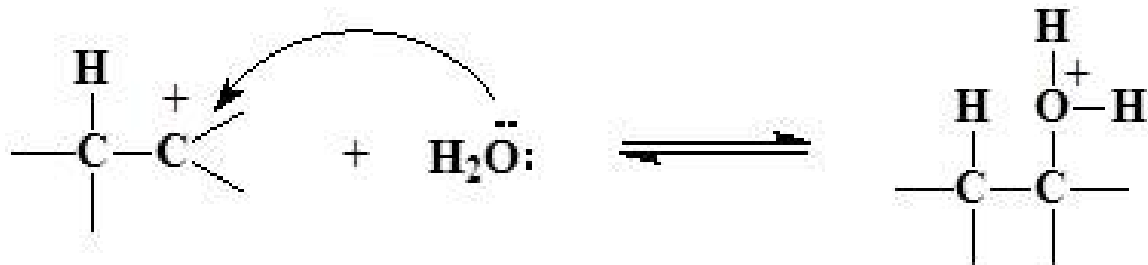
Hydration of Alkenes



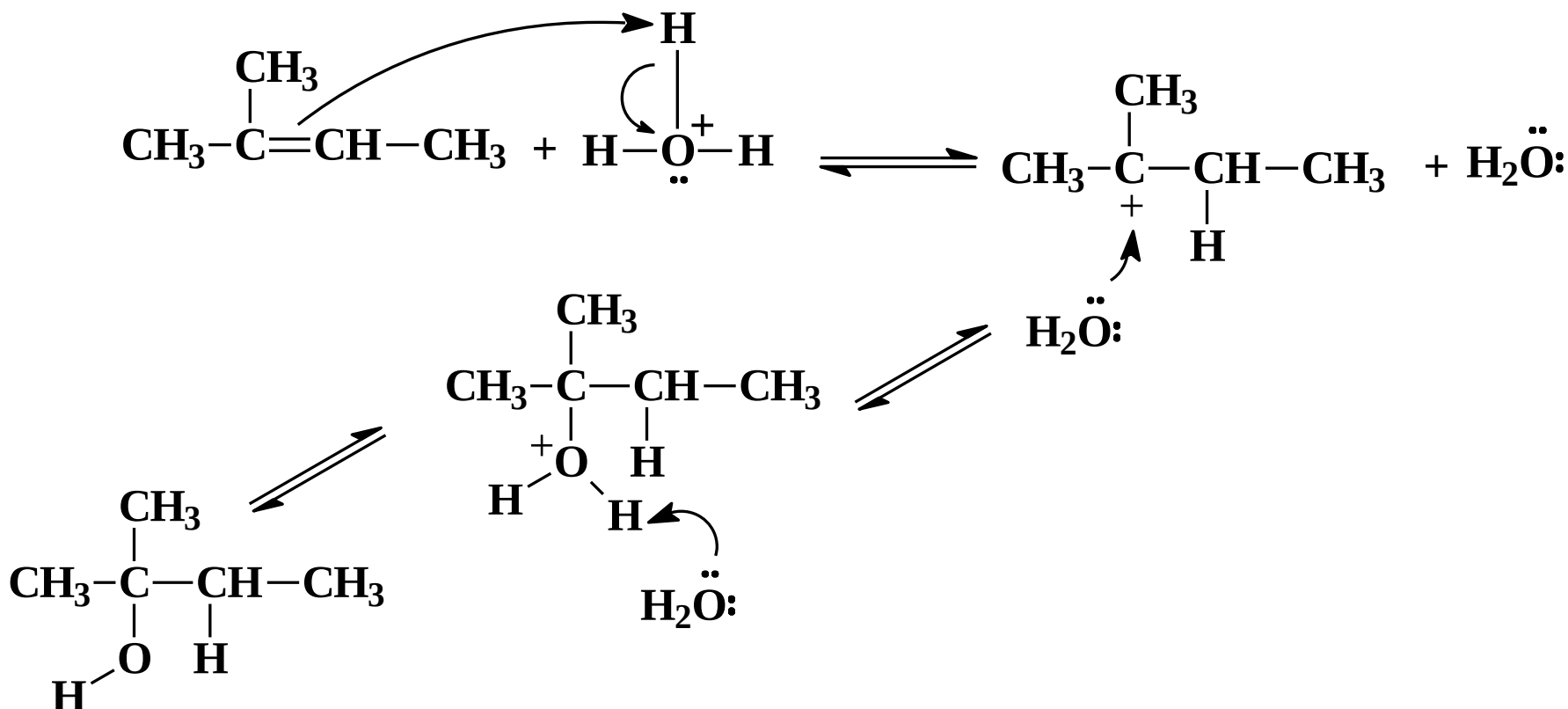
But: The O-H bond are too strong and not acidic enough to initiate The electrophilic addition

- It required an acid (H^+) to catalyze the reaction like H_2SO_4

Mechanism of Acid catalyzed hydration reaction

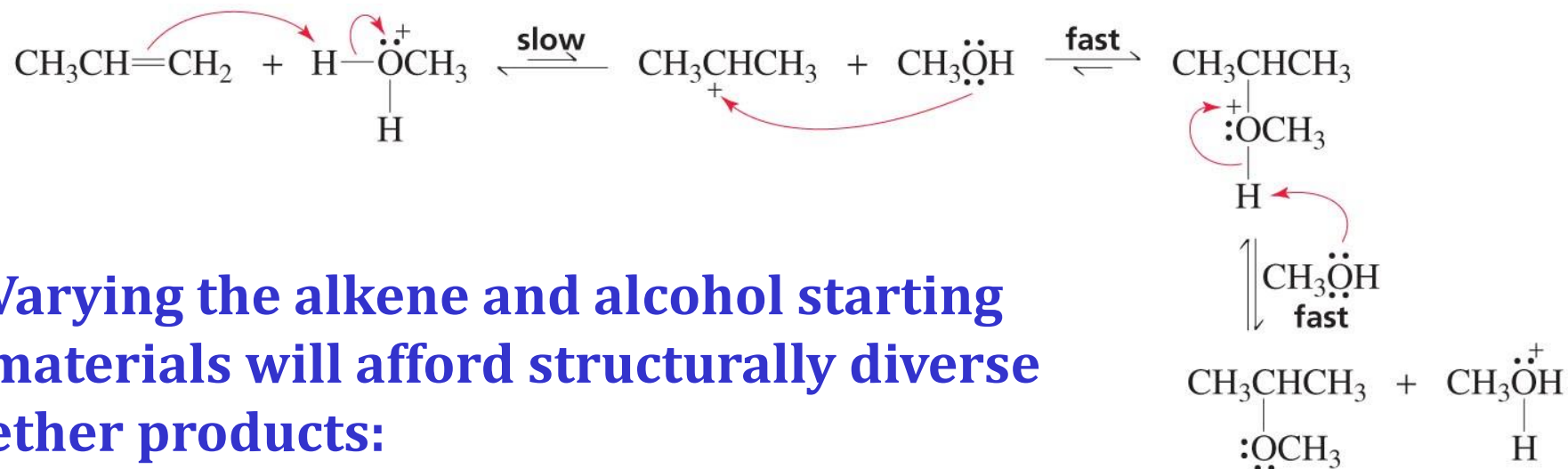


Mechanism of Acid catalyzed hydration reaction

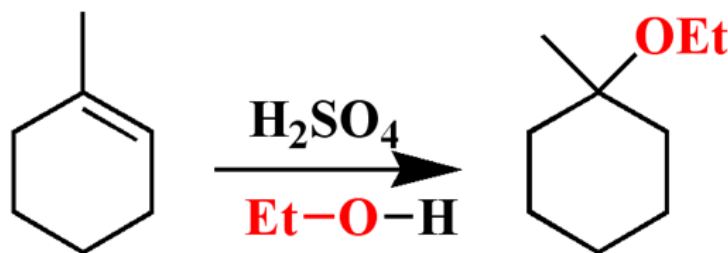


Acid Catalyzed Addition of Alcohol

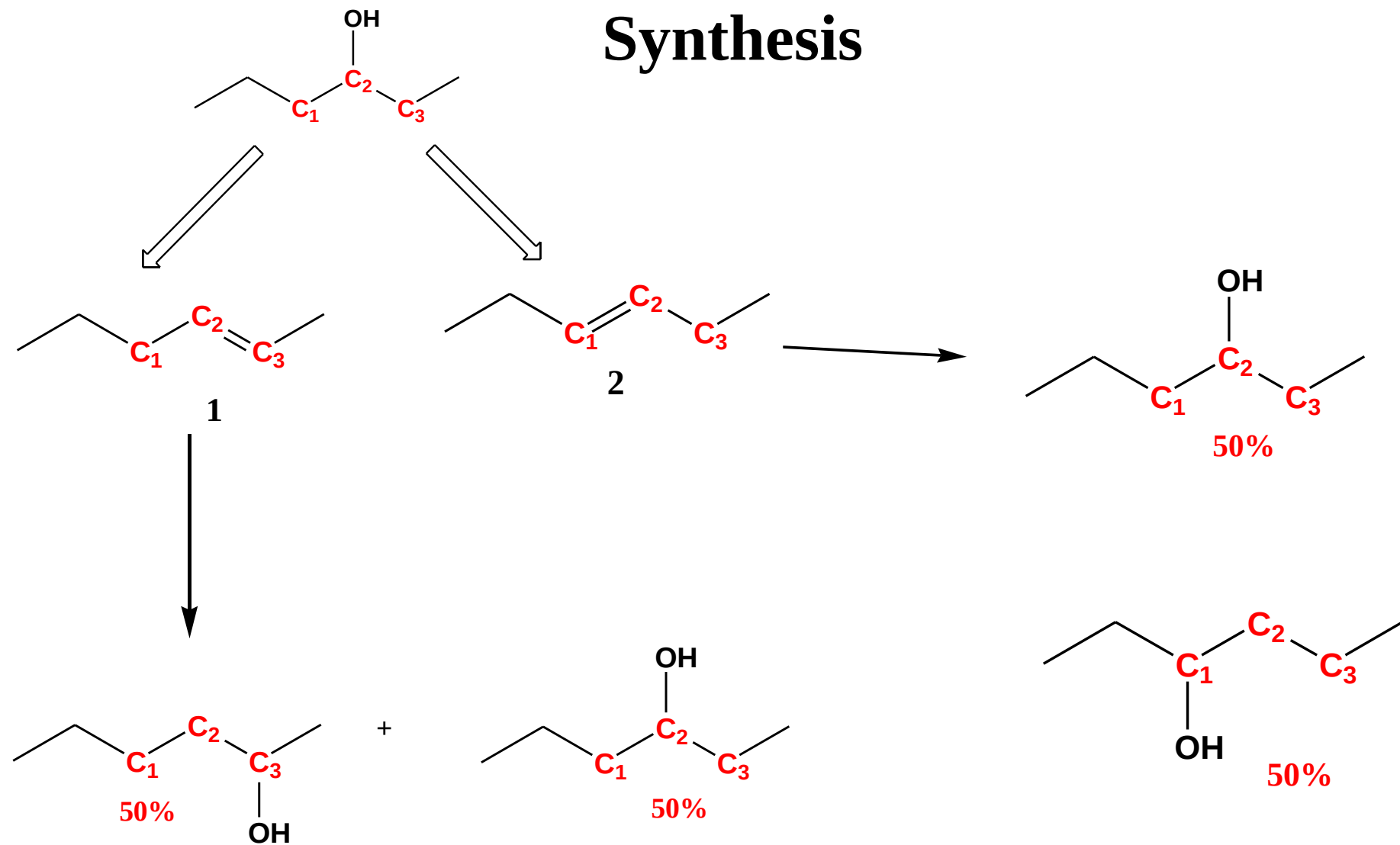
Mechanism:



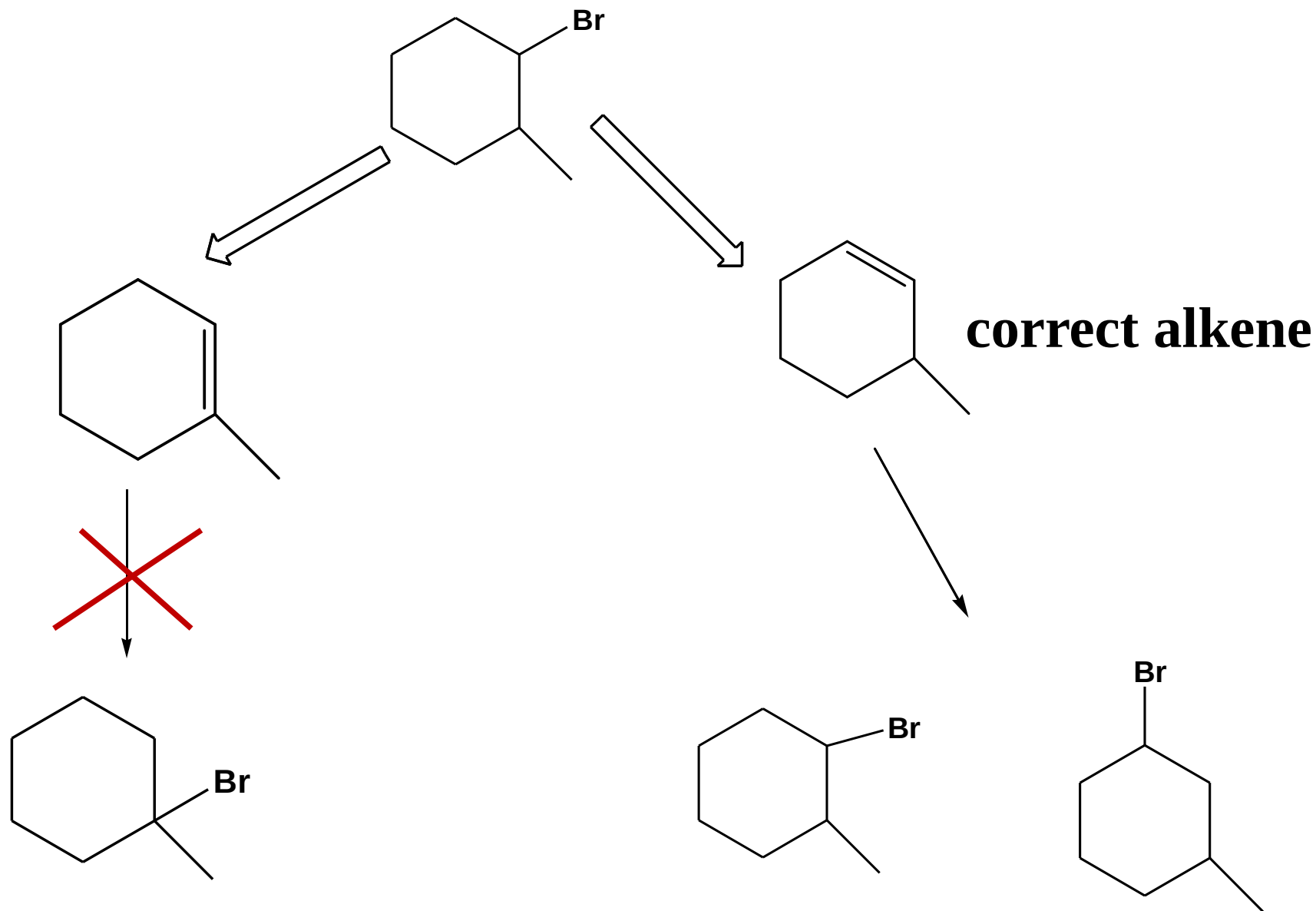
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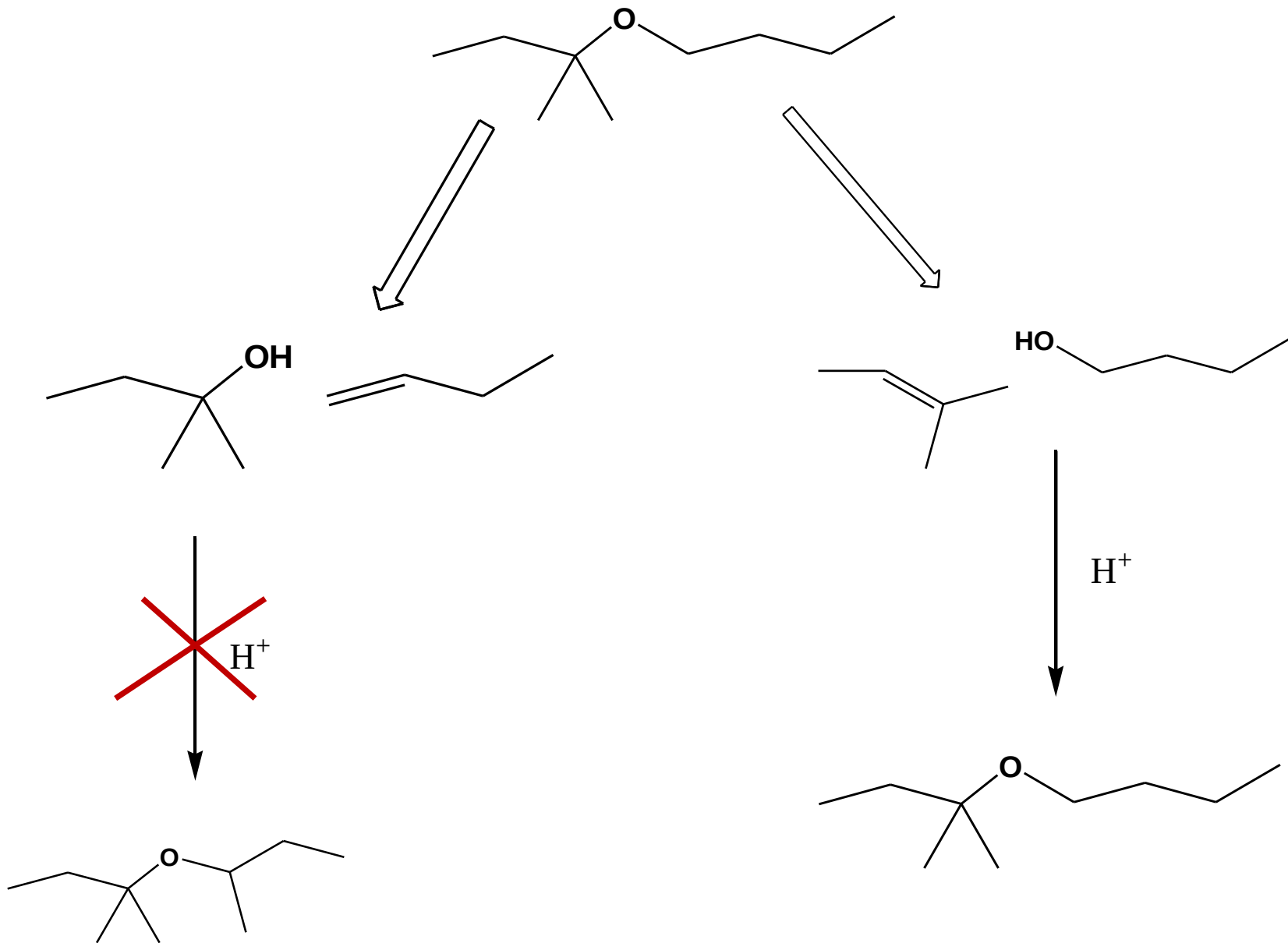


Synthesis



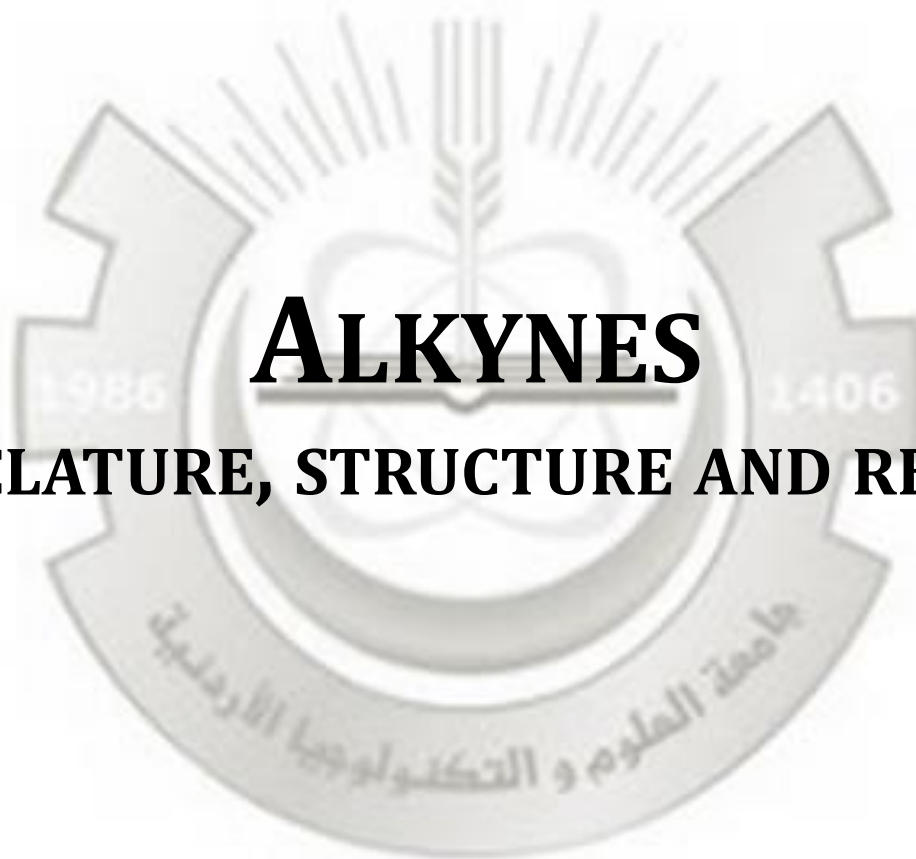
Alkene 2 is better choice





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ALKYNES

NOMENCLATURE, STRUCTURE AND REACTIVITY

Nomenclature

In common nomenclature, alkynes are named as substituted acetylenes:



Systematic:

ethyne

Common:

acetylene



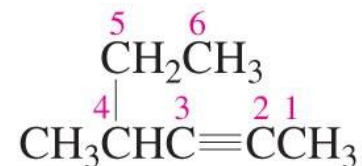
1-butyne

ethylacetylene
a terminal alkyne



2-pentyne

ethylmethylacetylene
an internal alkyne



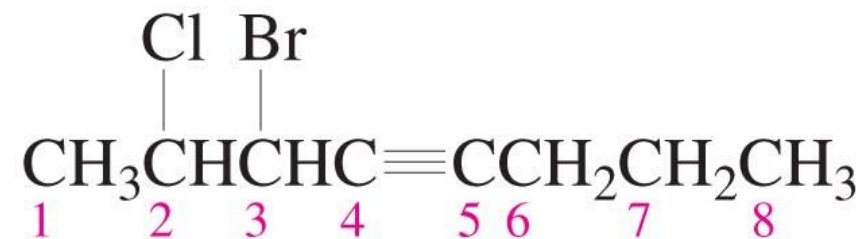
4-methyl-2-hexyne

sec-butylmethyl-
acetylene



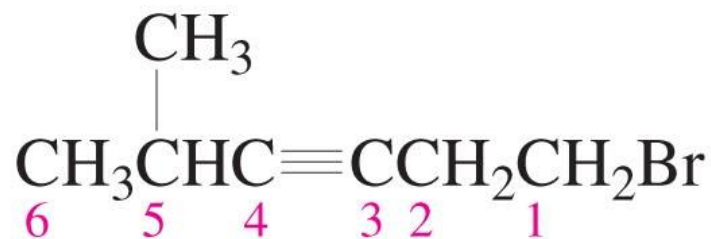
In IUPAC nomenclature:

- The yne suffix is assigned the lowest number.
- Then substituents are assigned the lowest possible set of numbers.



3-bromo-2-chloro-4-octyne

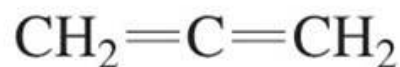
not 6-bromo-7-chloro-4-octyne
because $2 < 6$



1-bromo-5-methyl-3-hexyne

not 6-bromo-2-methyl-3-hexyne
because $1 < 2$

A compound containing two double bonds or two triple bonds:



propadiene

allene



2-methyl-1,4-hexadiene

or

2-methylhexa-1,4-diene



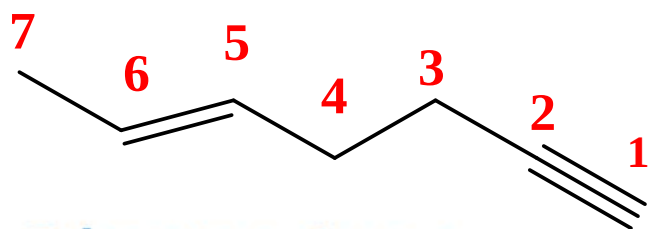
6-methyl-1,4-heptadiyne

or

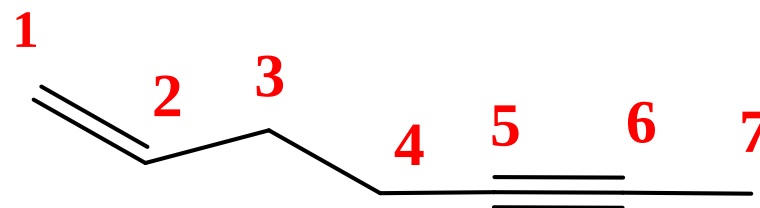
6-methylhepta-1,4-diyne

Alkene and Alkyne Mixed Functional Groups

- Find the longest chain bearing both functional groups.
- Assign ene and yne suffixes the lowest set of numbers.

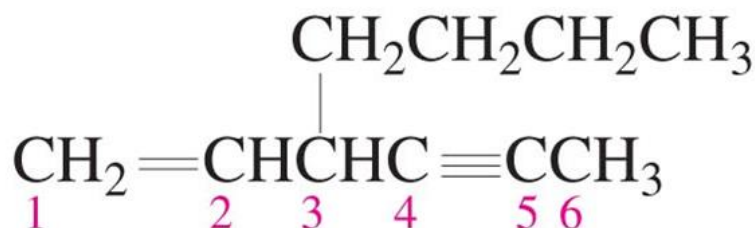


5-hepten-1-yne



1-hepten-5-yne

Longest chain
with both functional
groups



3-butyl-1-hexen-4-yne

the **ene** suffix is assigned the lower number:

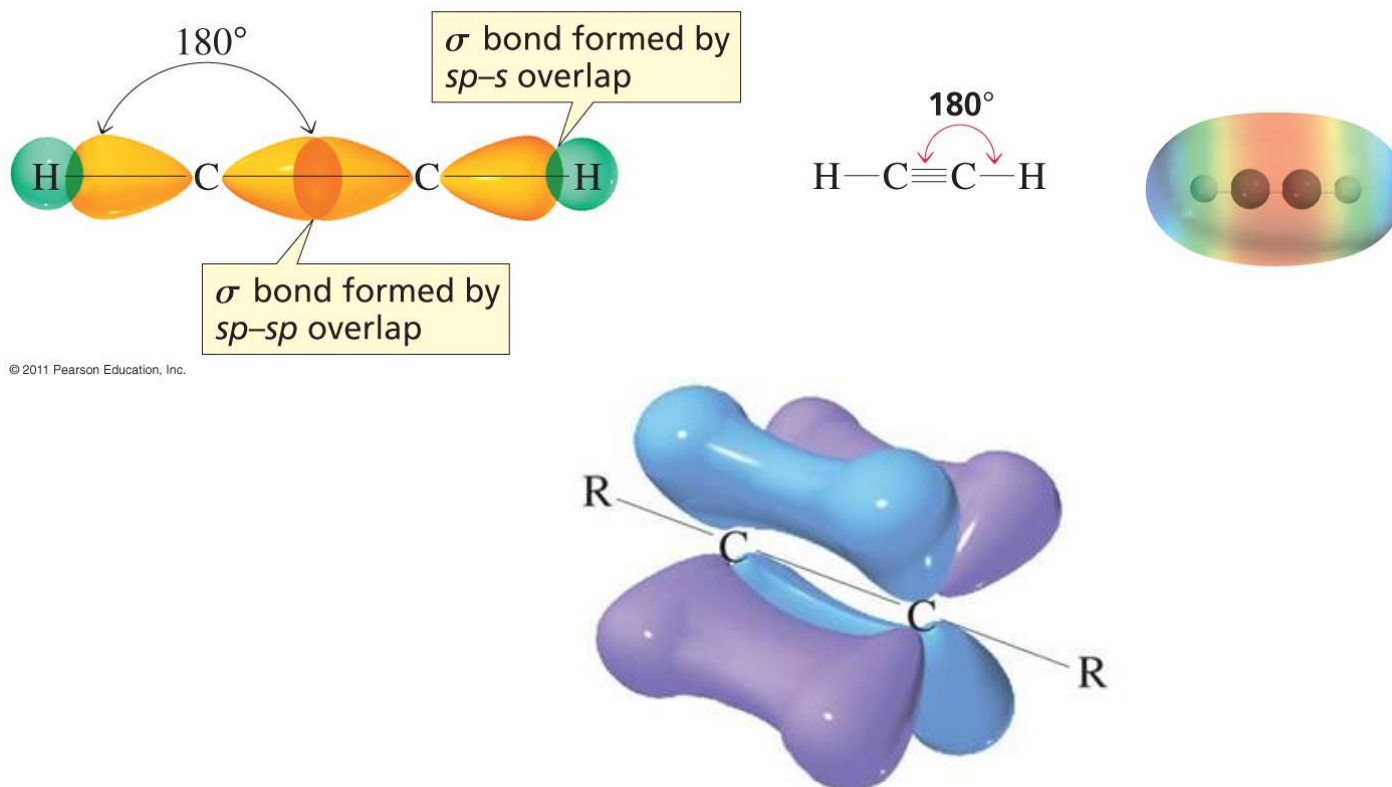


2-hexen-4-yne



1-hexen-5-yne

The Structure of Alkynes

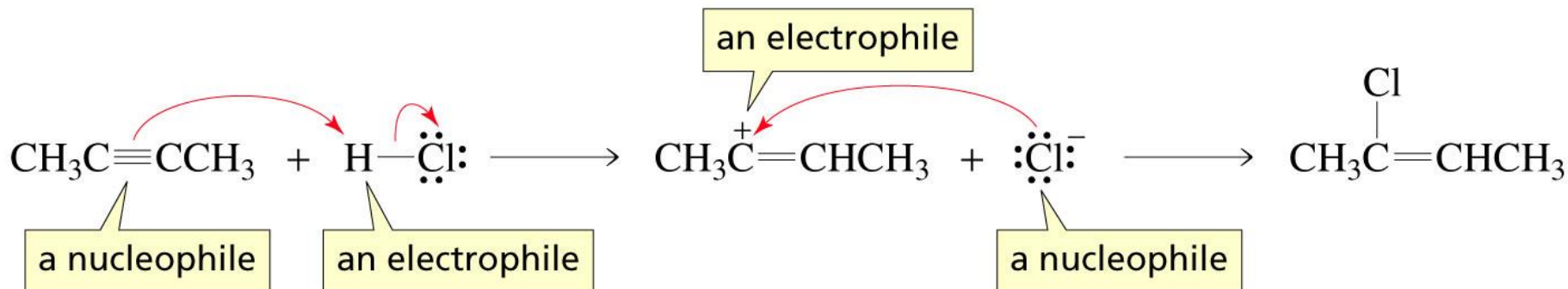


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A triple bond is composed of a σ bond and two π bonds

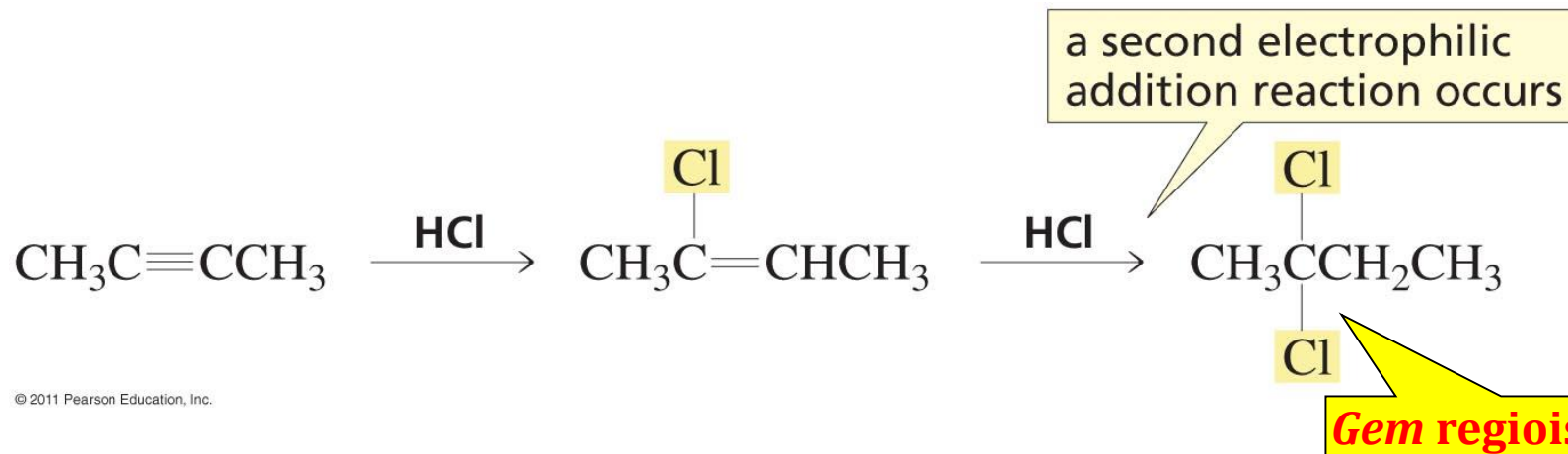
Electrophilic Addition of Hydrogen Halides to Alkynes

One Equivalent of HCl:



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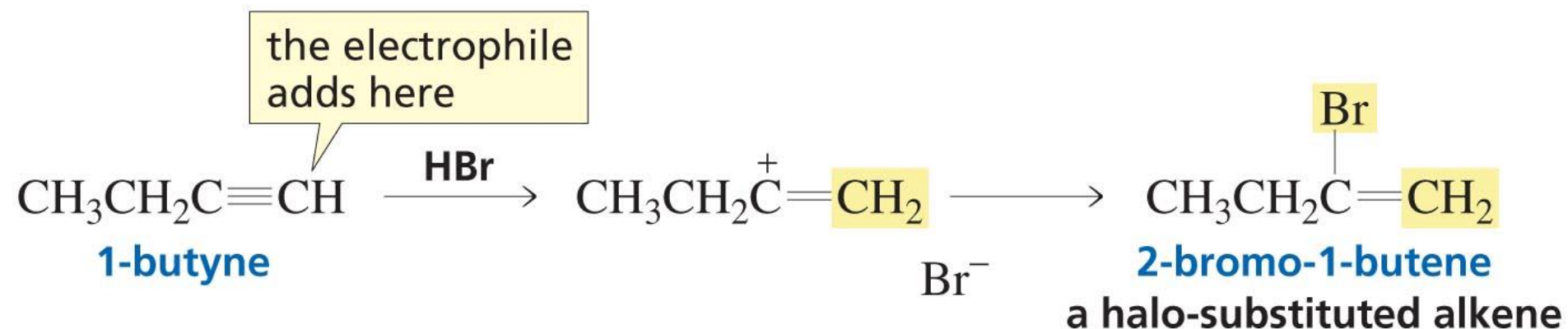
Excess HCl:



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Regioselectivity of Alkyne Addition Reactions

Use Markovnikov's rule for HCl and HBr additions to terminal alkynes:



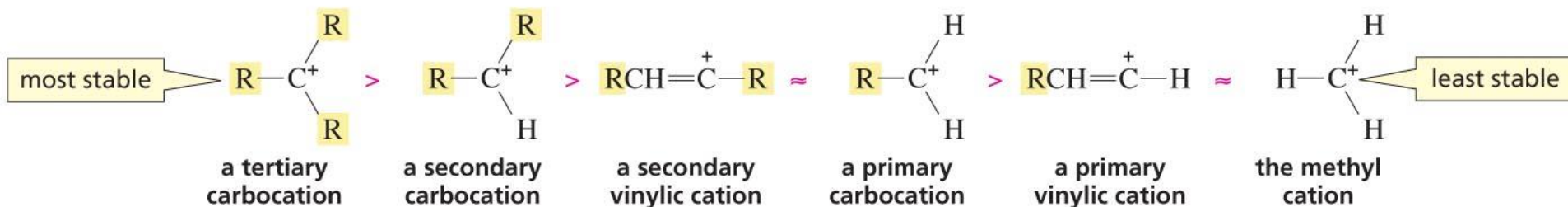
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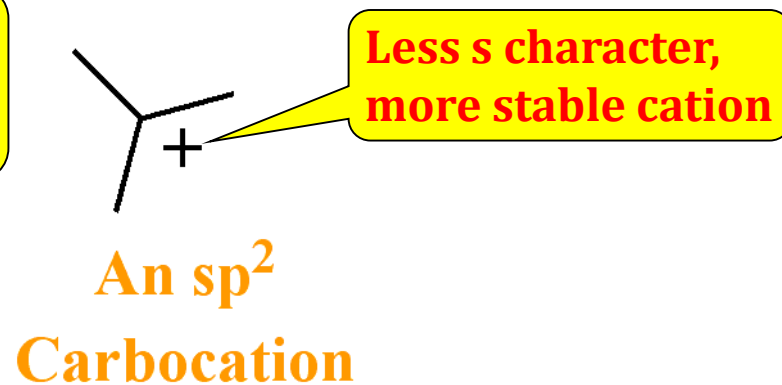
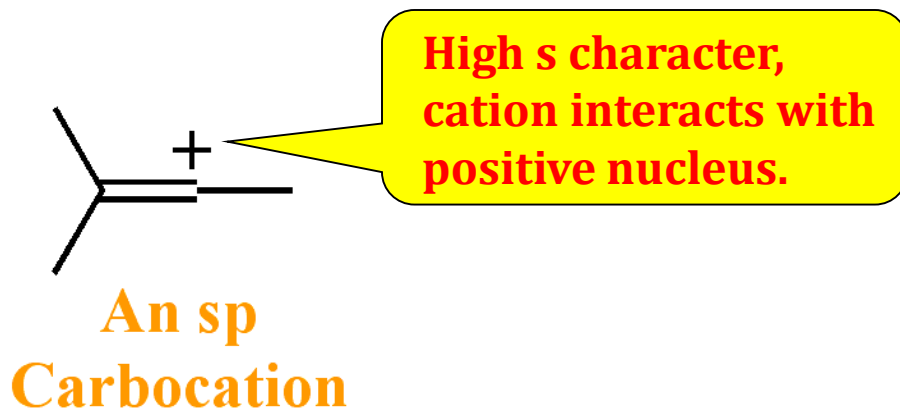
Relative Stabilities of Carbocations

relative stabilities of carbocations

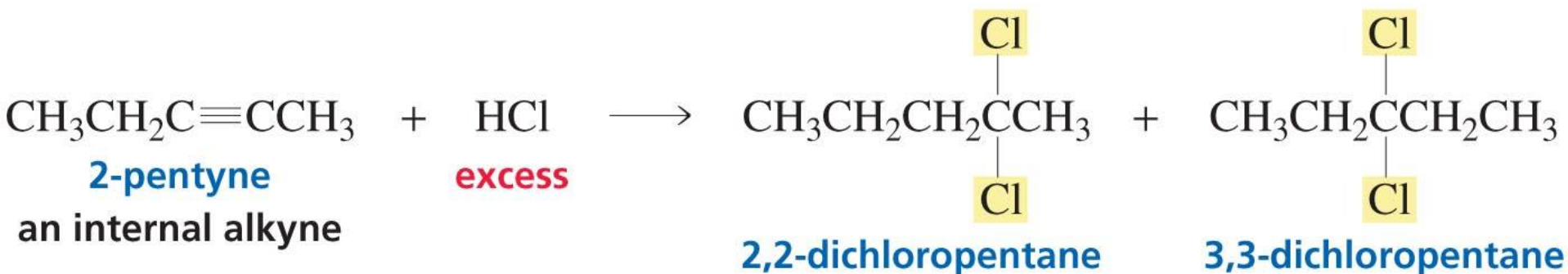


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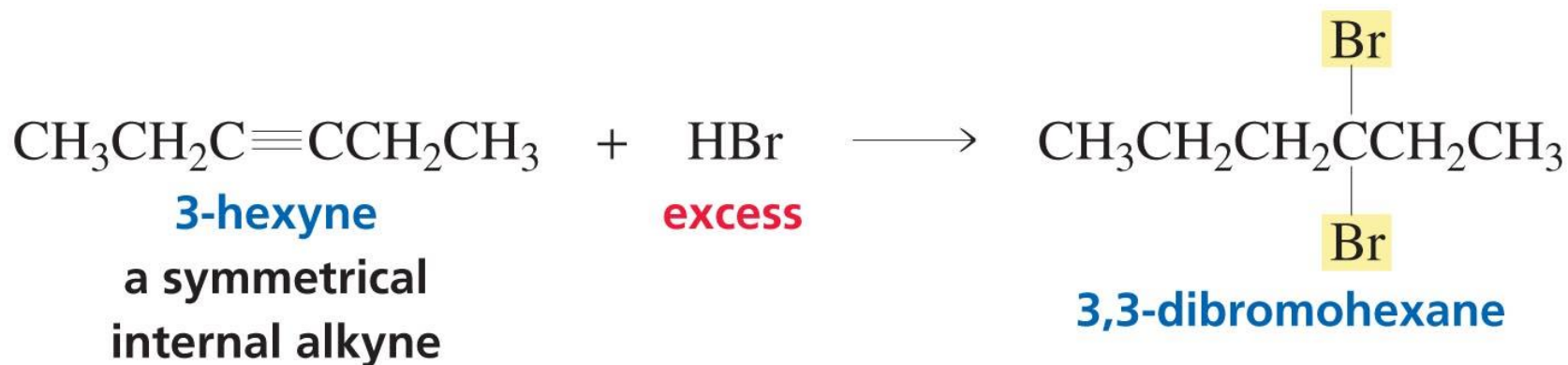
A vinylic cation (sp) is formed in the first step of the alkyne addition reaction whereas an alkyl cation (sp^2) is formed in the first step of the alkene addition reaction



The initial addition of the proton can occur with equal ease to either of the *sp* carbons and the ***geminal*** regioisomer always results:

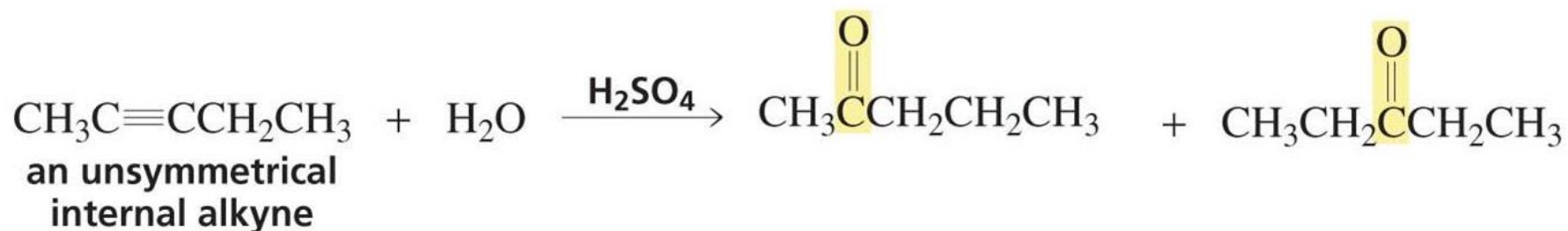
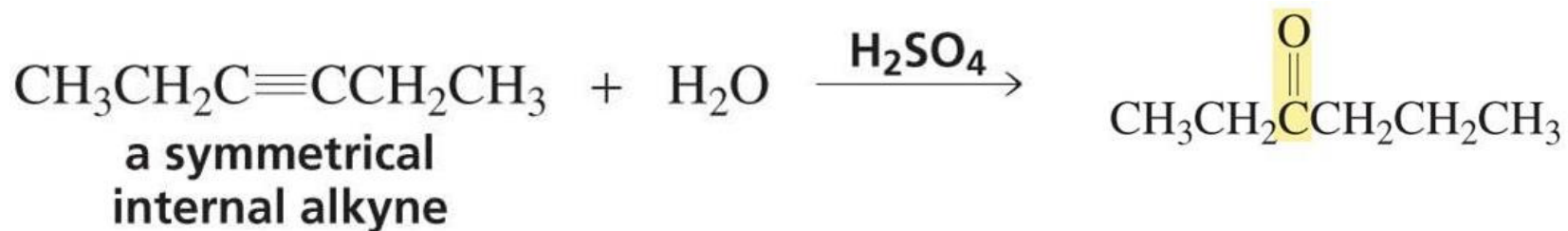
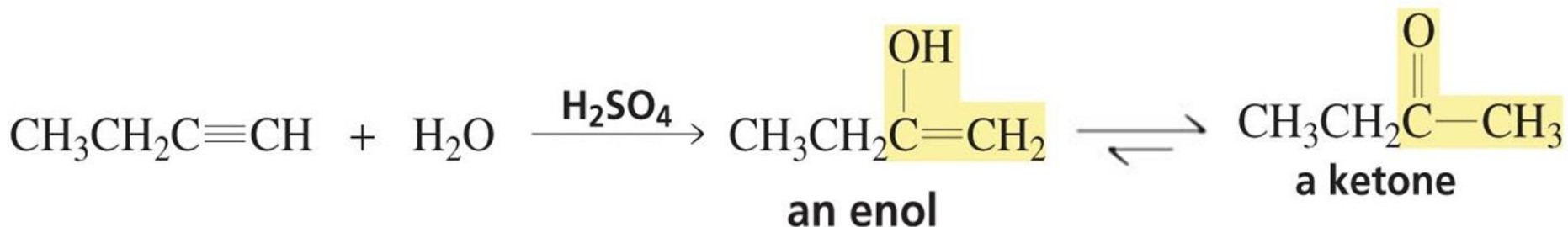


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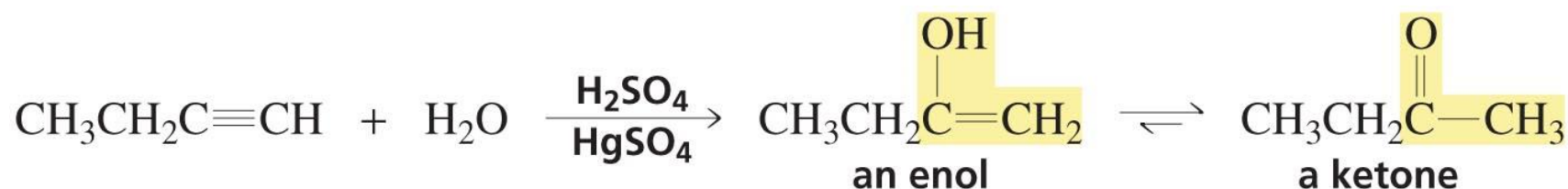
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Addition of Water to an Alkyne



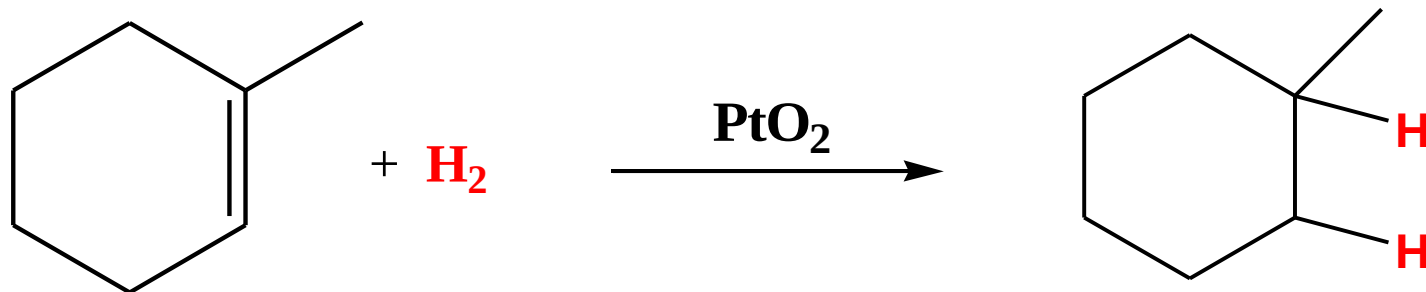
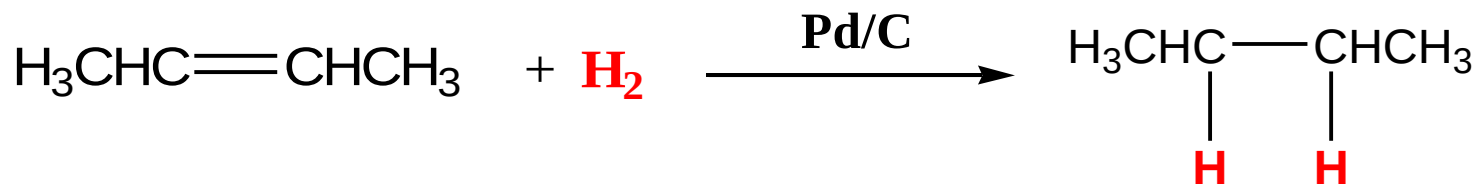
One method of synthesizing ketones.

Hg^{2+} is added to increase the rate of water addition to terminal alkynes:

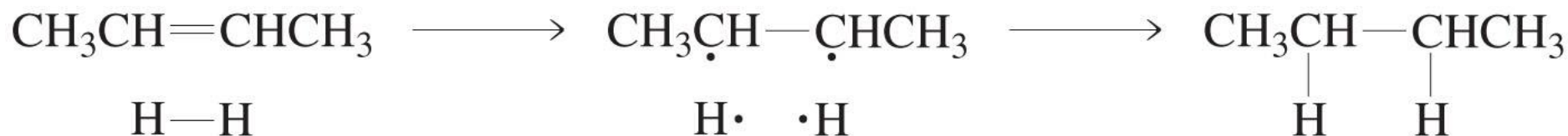
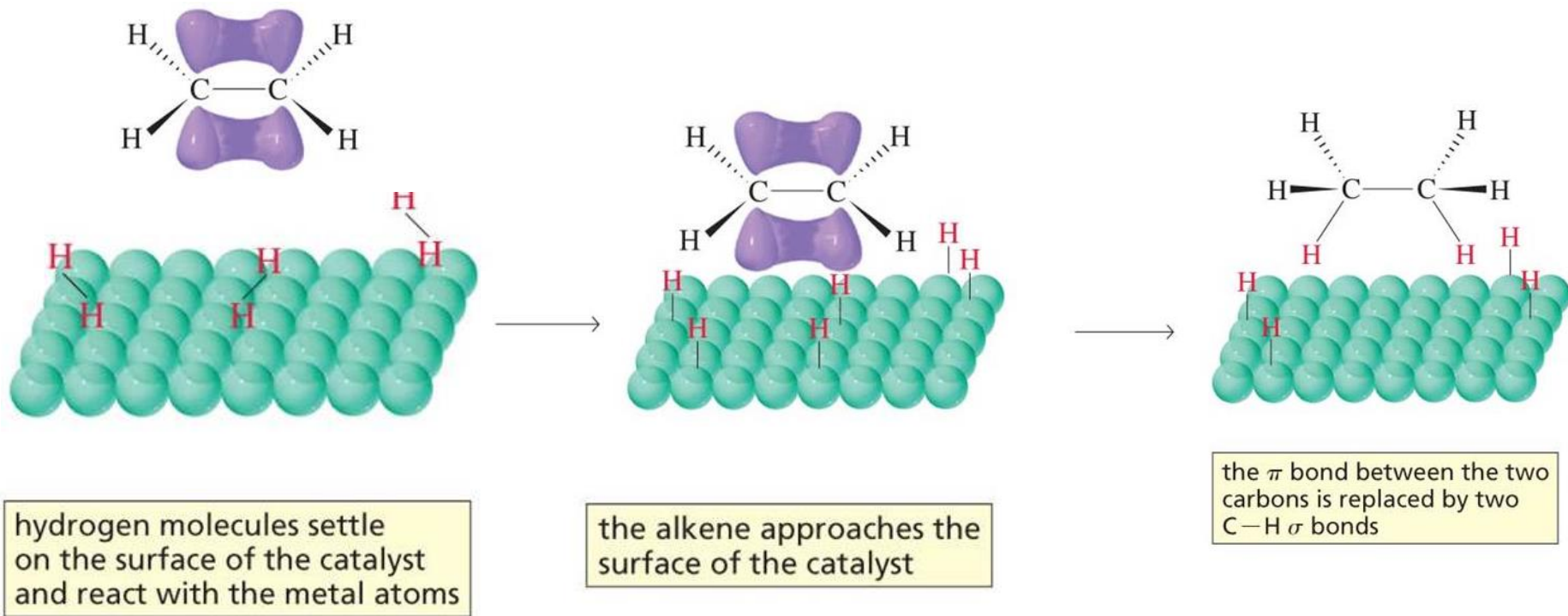


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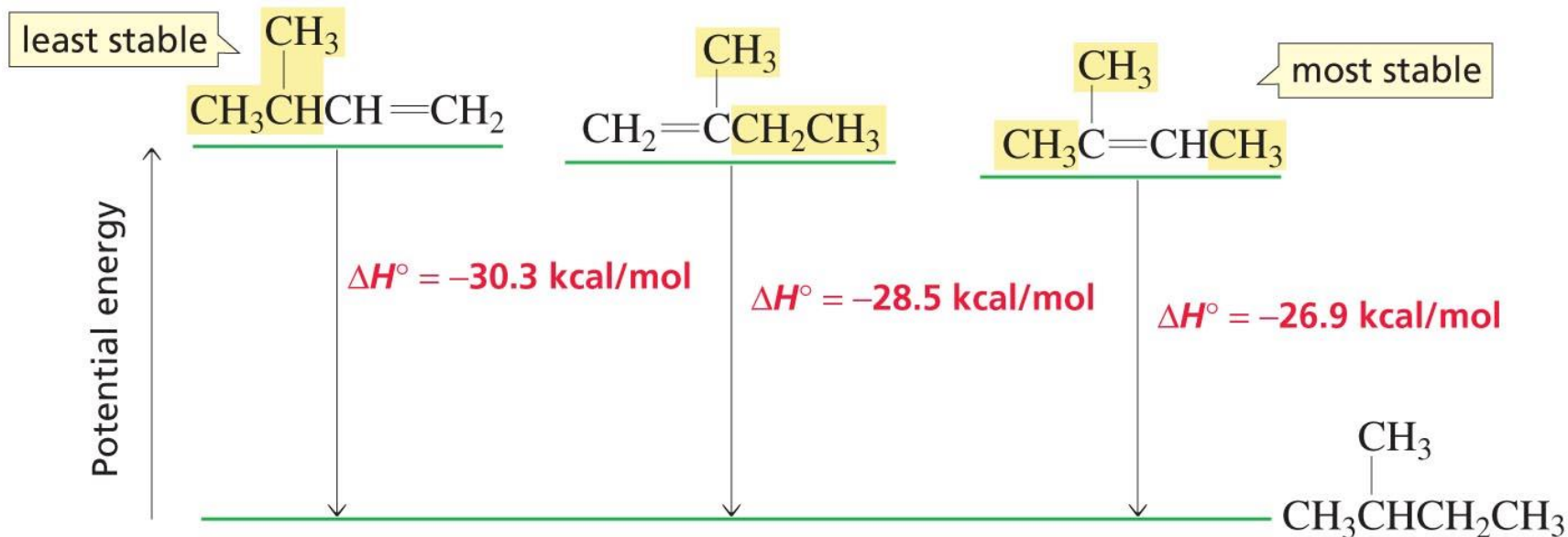
Addition of Hydrogen to Alkenes



Catalytic Hydrogenation of Alkenes



Potential Energies of Pentene Isomers

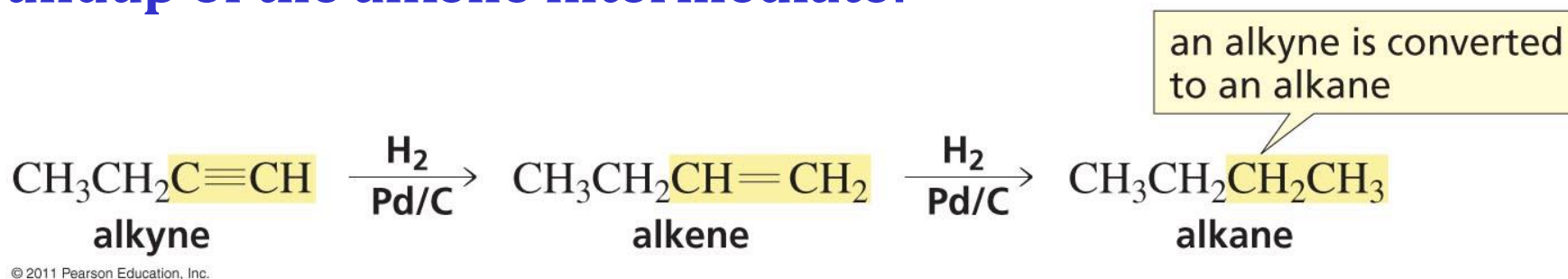


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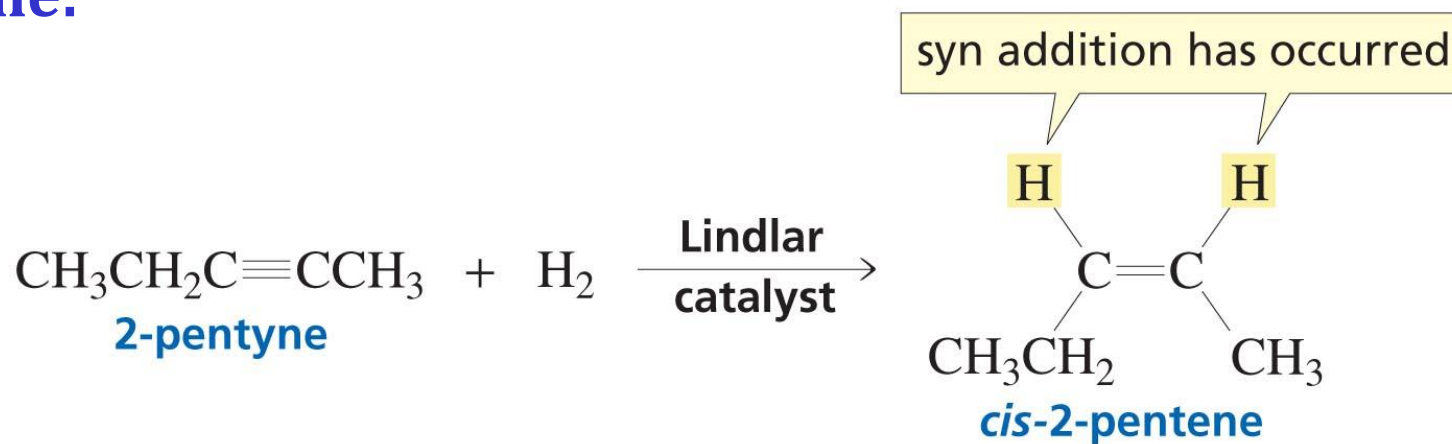
The most stable alkene has the smallest heat of hydrogenation.

Addition of Hydrogen Formation of Cis Alkene

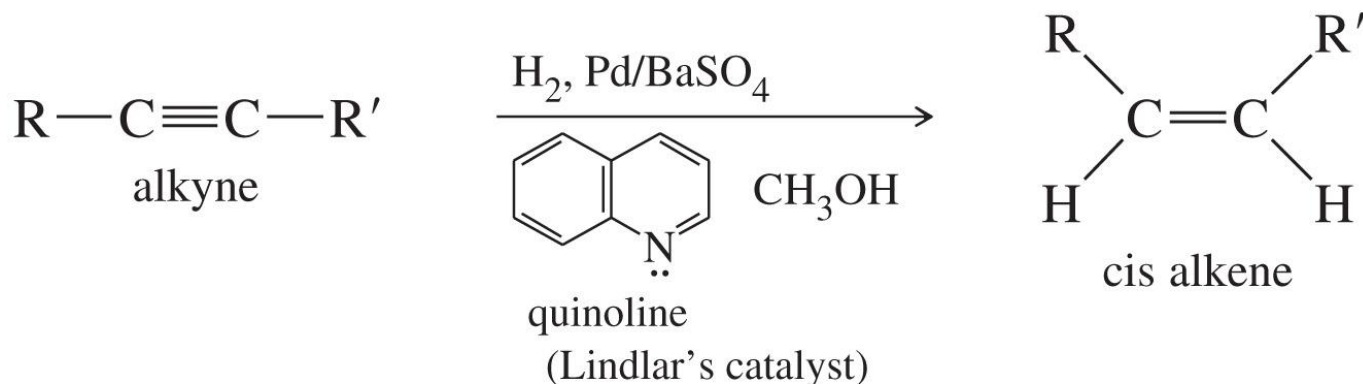
Catalytic reduction of an alkyne affords an alkane without buildup of the alkene intermediate:



Use a “poisoned” catalyst developed by Lindlar to obtain the alkene:



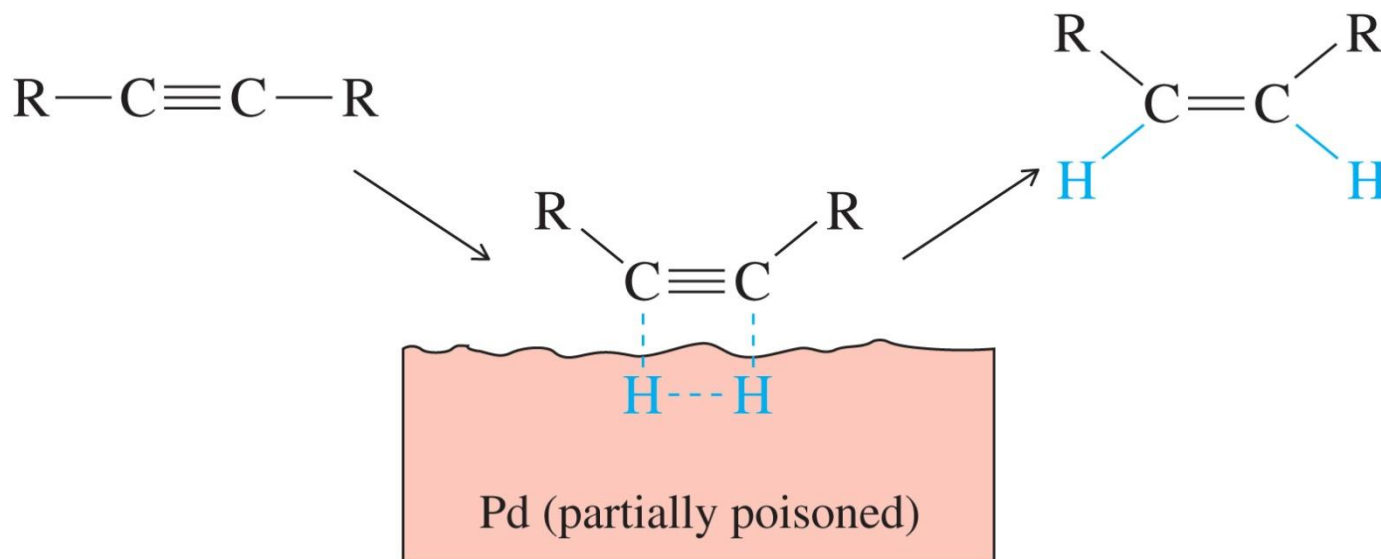
Hydrogenation with Lindlar's Catalyst



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- The catalyst used for the hydrogenation reaction is partially deactivated (poisoned), the reaction can be stopped after the addition of only one mole of hydrogen.
- The catalyst used is commonly known as Lindlar's catalyst and it is composed of powdered barium sulfate, coated with palladium poisoned with quinoline.
- The reaction produces alkenes with cis stereochemistry.

Mechanism



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- Both substrates, the hydrogen and the alkyne, have to be adsorbed on the catalyst for the reaction to occur.
- Once adsorbed, the hydrogens add to the same side of the double bond (syn addition) giving the product a cis stereochemistry.

Acidity of a Hydrogen Bonded to an *sp* Hybridized Carbon

relative electronegativities of carbon atoms

most
electronegative

sp

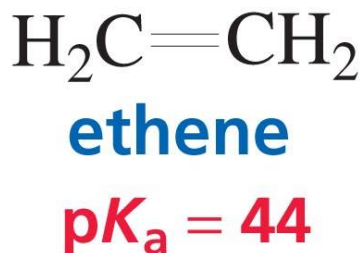
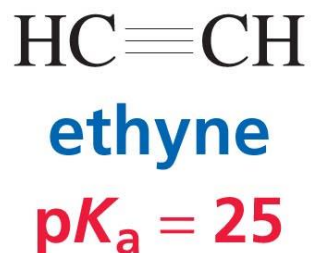
High *s* character, *sp*
orbital penetrates
nucleus.

$> sp^2$

$> sp^3$

least
electronegative

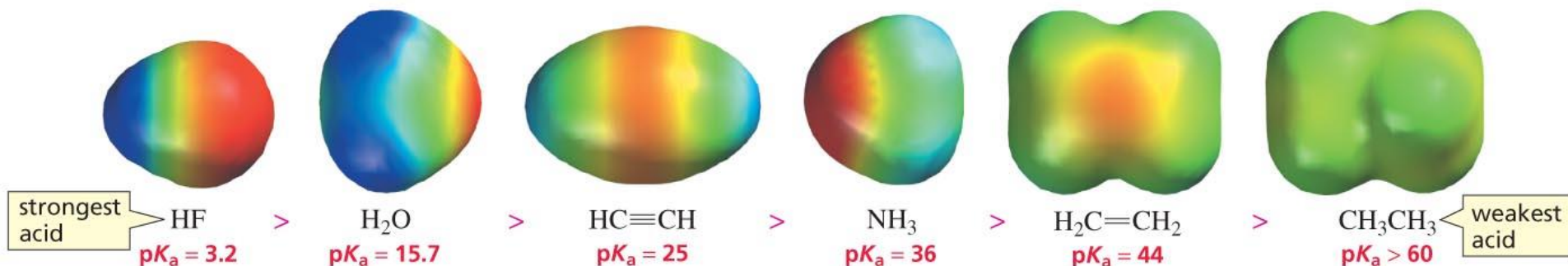
Lower *s* character,
*sp*³ orbital
penetrates nucleus
to a lesser degree



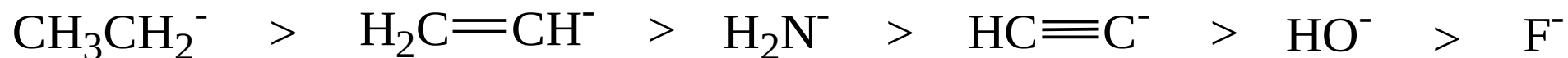
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Electronegativities versus Acid Strengths

relative acid strengths



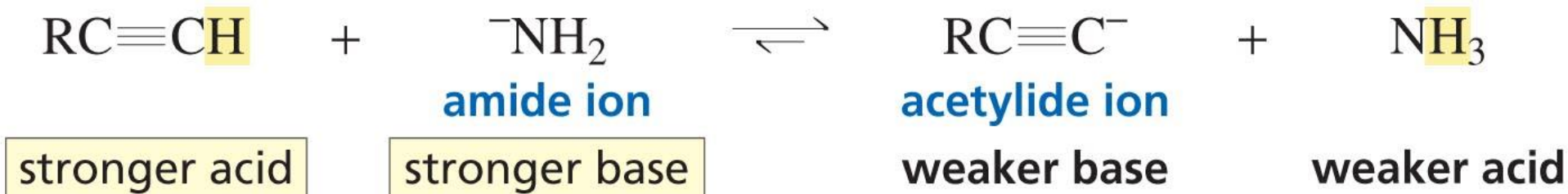
relative base strength



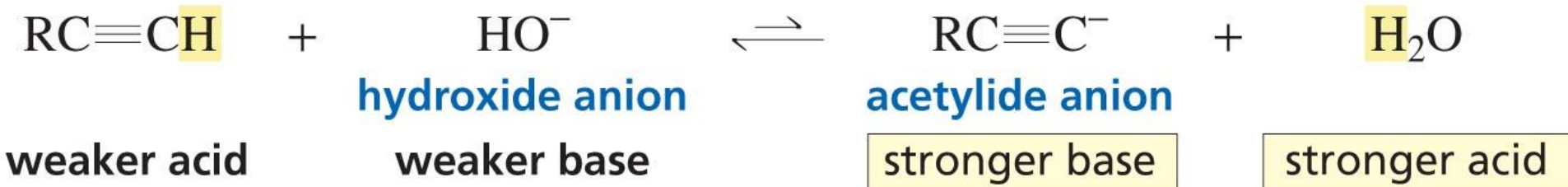
strongest
base

weakest
base

The stronger the acid, the weaker its conjugate base:



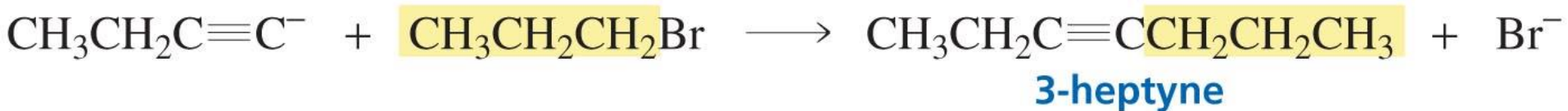
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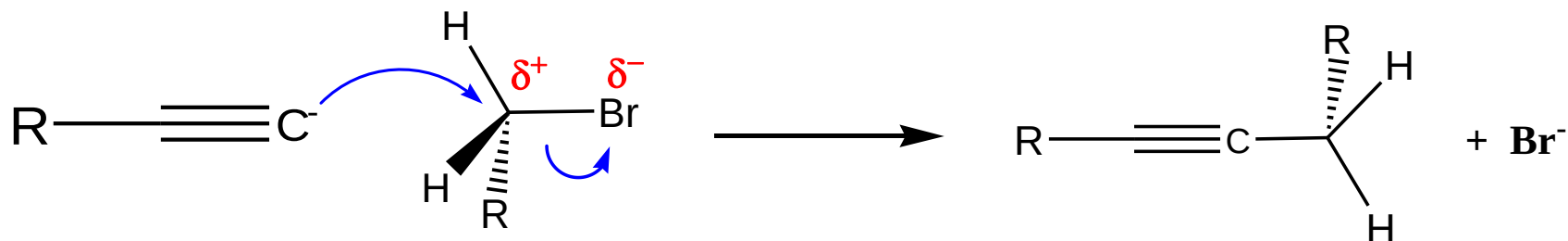
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An amide anion cannot remove a hydrogen bonded to an sp^2 or sp^3 carbon.

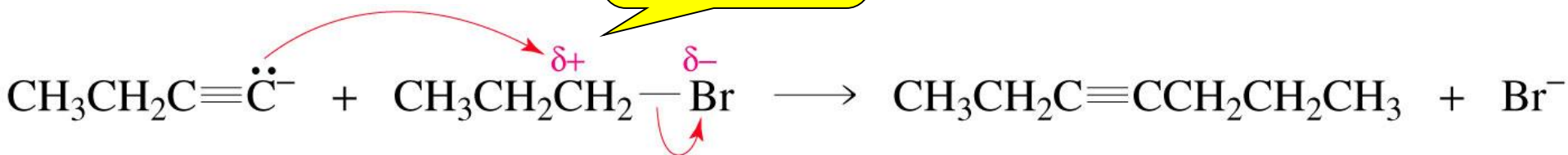
Synthesis using Acetylide ions: formation of C–C Bond



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Known as $\text{S}_{\text{N}}2$ reaction

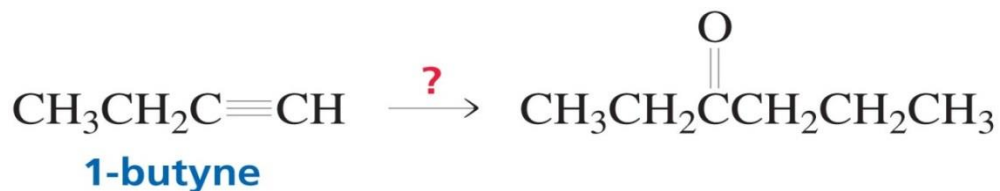


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We can convert terminal alkynes to longer internal alkynes by addition of alkyl halides

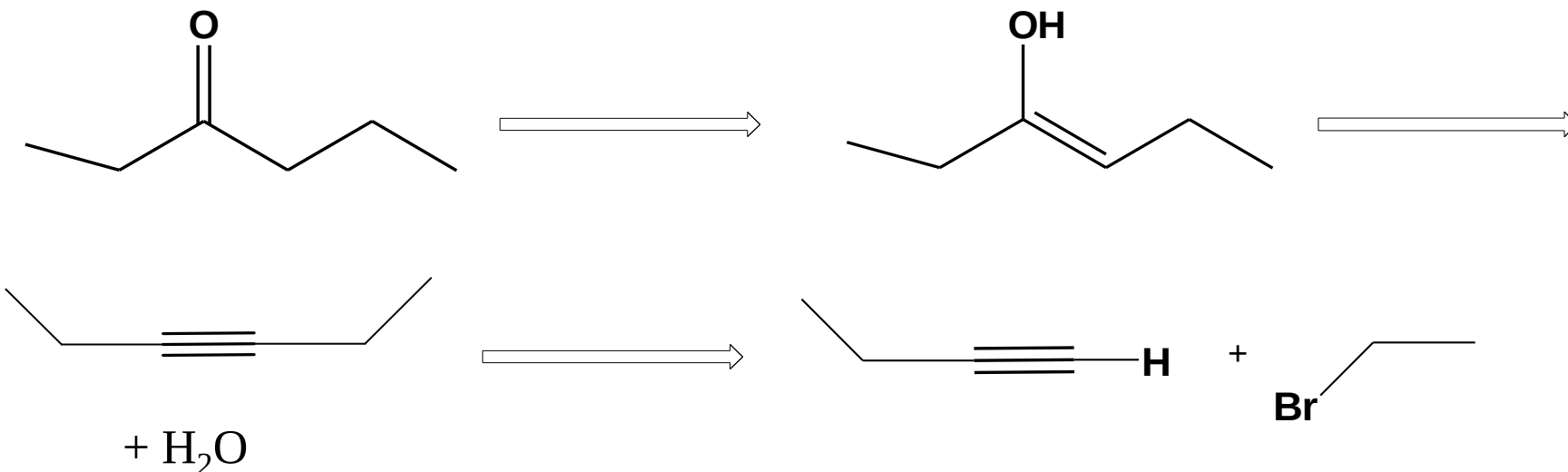
Designing a Synthesis

Example 1

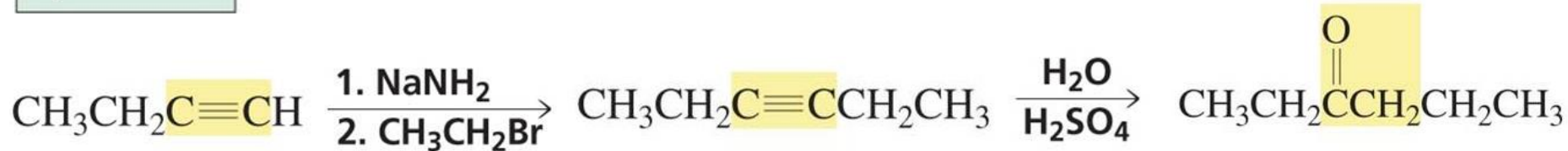


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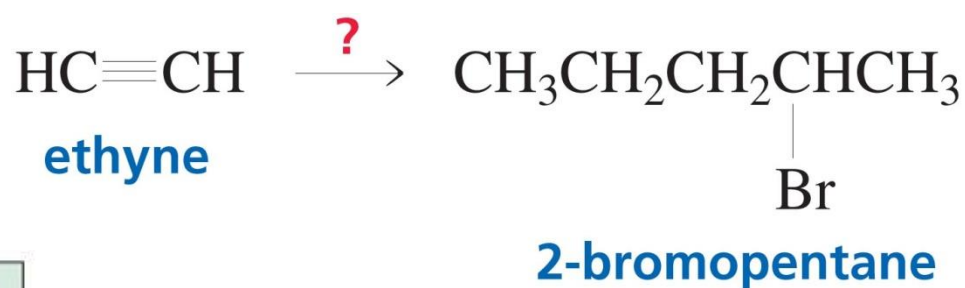
Retrosynthesis



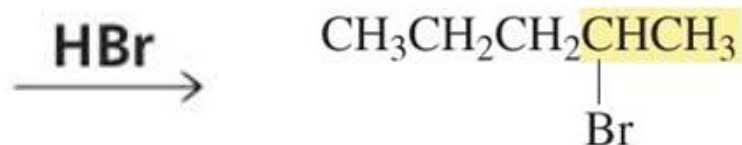
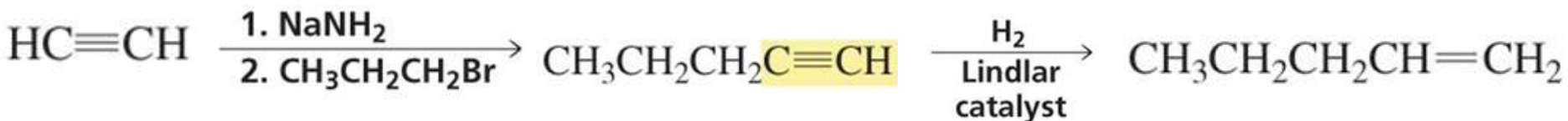
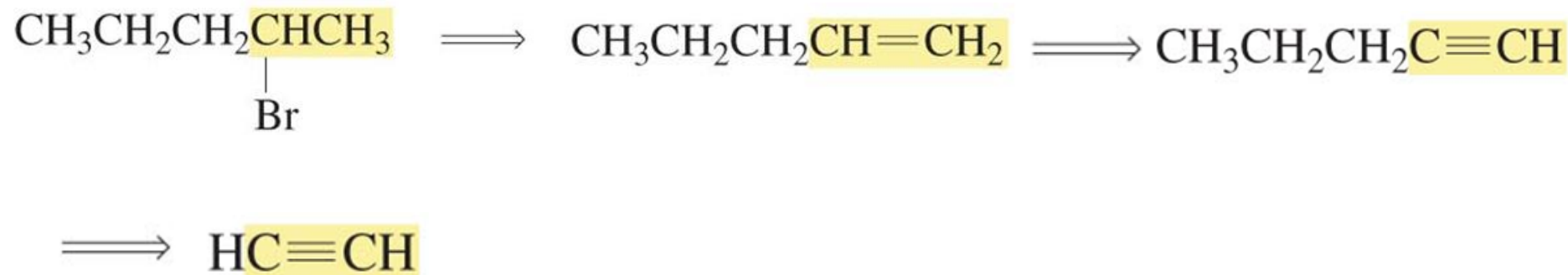
synthesis



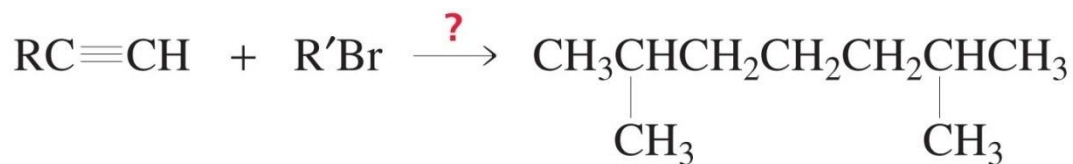
Example 2



retrosynthetic analysis

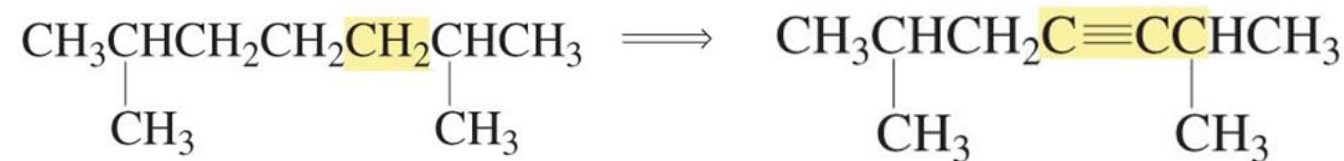


Example 3

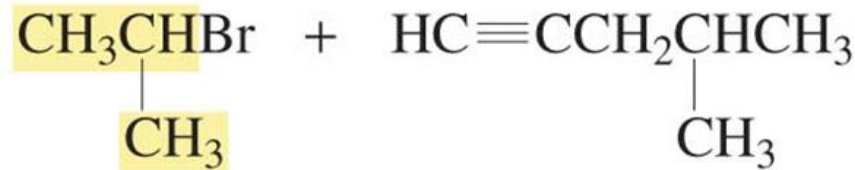
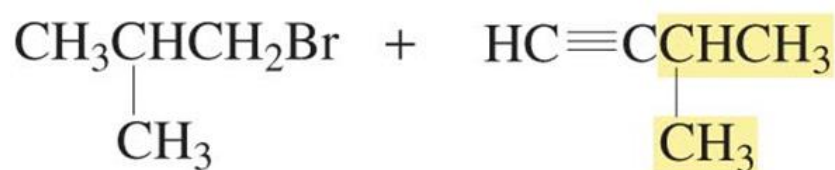


retrosynthetic analysis

2,6-dimethylheptane



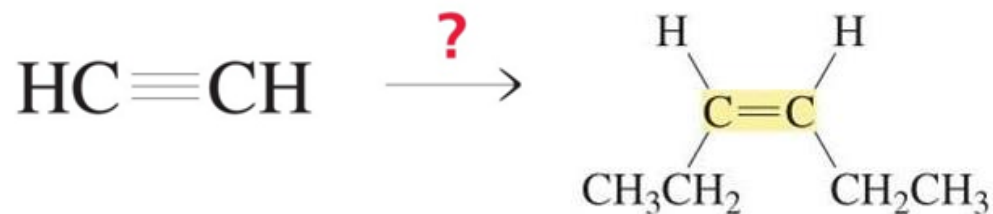
Secondary alkyl halide



synthesis



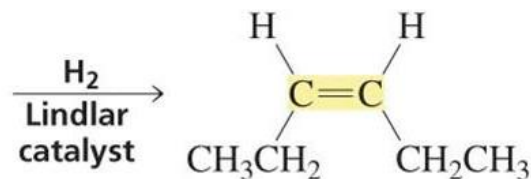
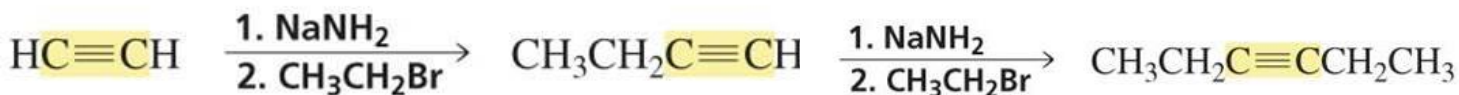
Example 4



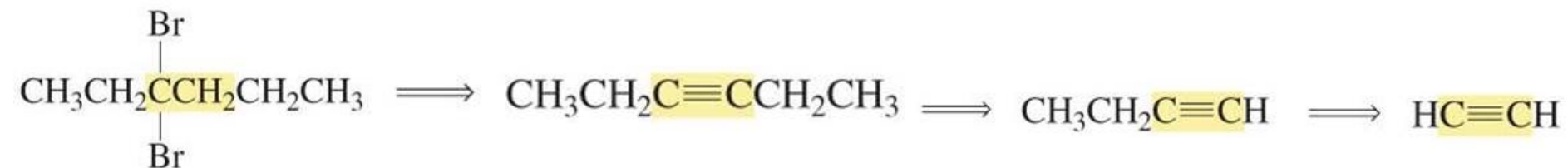
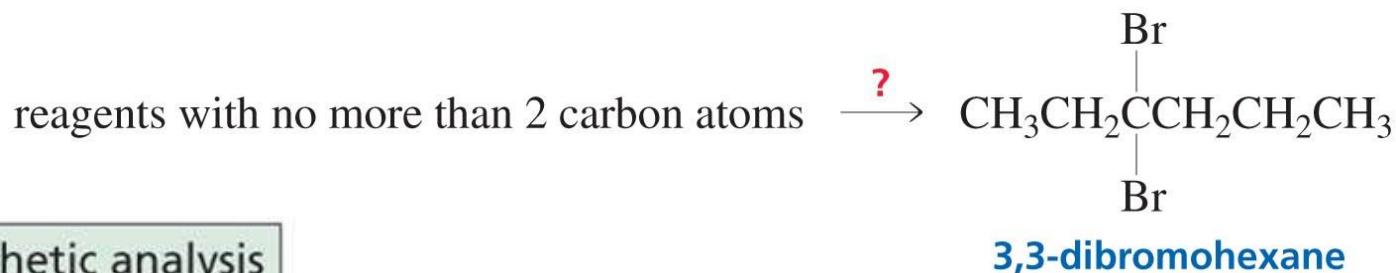
retrosynthetic analysis



synthesis



Example 5



synthesis

